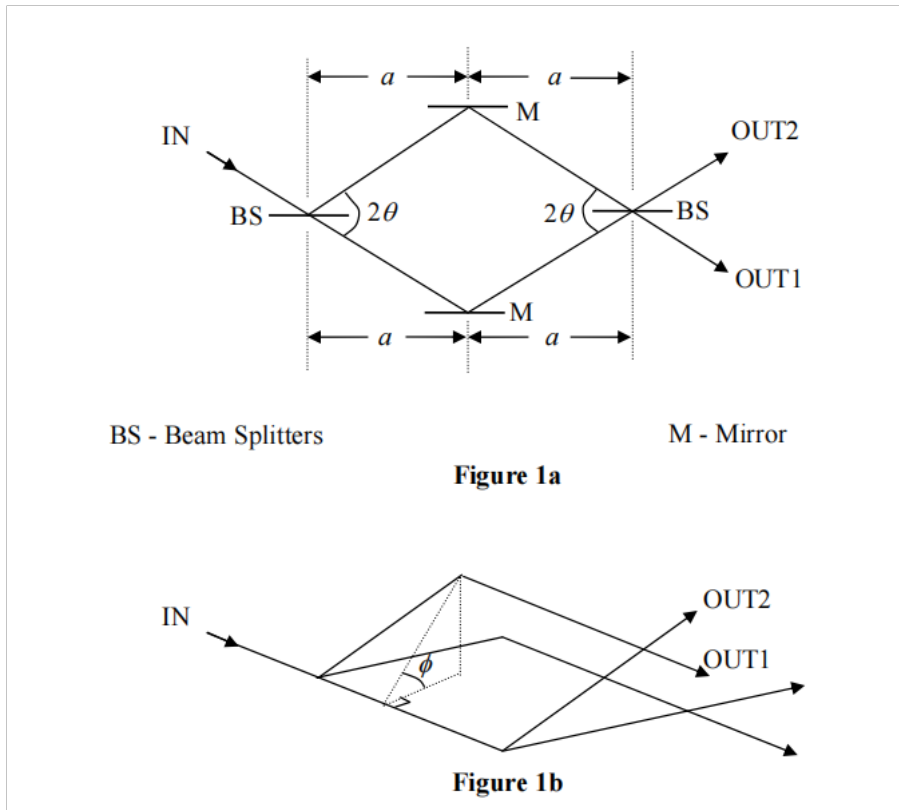


Nazariy masala 1: Neytron interferometrida tortishish (Gravity in a Neutron Interferometer)



1a-rasm:

Interferometrning simvolli tasviri, 1b-rasm: Interferometrning ϕ burchakka og'ishi.

Fizik holat (Physical situation)

Biz Kollera, Overhauser and Verner tajribasidagi mashhur neytron-interferometr holatini ko'rib chiqamiz, lekin interferometr ichida mukammal nur ajratgichlar (perfect beam splitters) va ko'zgular (mirrors) bor deb idealizatsiya qilamiz. Tajriba gravitatsion tortishning neytronlarning de Broyl to'lqinlariga (de Broglie waves) ta'sirini o'rganadi.

Interferometrning optik interferometrga o'xshash simvolli tasviri 1a-rasmda ko'rsatilgan. Neytronlar interferometrga IN kirish porti (port) orqali kiradi va ko'rsatilgan ikki yo'lni (paths) bosib o'tadi. Neytronlar ikkita chiqish portidan (OUT1 yoki OUT2) birida aniqlanadi. Ikki yo'l odatda bir necha sm^2 kattalikdagi olmos shaklidagi maydonni (diamond-shaped area) o'rab oladi.

De Broyl to'lqinlari (odatdagi to'lqin uzunligi 10^{-10} m) shunday interferensiyalanadiki, agar interferometr tekisligi gorizontal bo'lsa, barcha neytronlar OUT1 chiqish portidan chiqadi. Ammo interferometr tushayotgan neytron nurlari o'qi atrofida ϕ burchakka (1b-rasm) og'dirilganda, neytronlarning ϕ ga bog'liq ravishda ikki chiqish porti OUT1 va OUT2 o'rtasida qayta taqsimlanishi kuzatiladi.

Geometriya (Geometry)

$\phi = 0^\circ$ da interferometr tekisligi gorizontal; $\phi = 90^\circ$ da tekislik vertikal bo'lib, chiqish portlari og'ish o'qidan (tilt axis) yuqorida joylashgan.

1.1 (1.0 ball) Interferometrning ikki yo'li o'rab olgan olmos shaklidagi maydon (area) A qancha?

1.2 (1.0 ball) OUT1 chiqish portining og'ish o'qining gorizontal tekisligidan balandligi H qancha?

A va H ni a , θ va ϕ orqali ifodalang.

Optik yo'l uzunligi (Optical path length)

Optik yo'l uzunligi N_{opt} (son) – geometrik yo'l uzunligi (masofa) ning to'lqin uzunligi λ ga nisbati. Agar λ yo'l bo'ylab o'zgarsa, N_{opt} λ^{-1} ni yo'l bo'ylab integrallash orqali olinadi.

1.3 (3.0 ball) Interferometr ϕ burchakka og'dirilganda ikki yo'lning optik yo'l uzunliklari farqi ΔN_{opt} qancha? Javobingizni a , θ , ϕ , shuningdek neytron massasi M , tushayotgan neytronlarning de Broyl to'lqin uzunligi λ_0 , gravitatsion tezlanish g va Plank doimiysi h orqali ifodalang.

1.4 (1.0 ball) Hajm parametrini (volume parameter) kiriting:

$$V = \frac{h^2}{gM^2}$$

va ΔN_{opt} ni faqat A , V , λ_0 va ϕ orqali ifodalang. $M = 1.675 \times 10^{-27}$ kg, $g = 9.800 \text{ m s}^{-2}$, $h = 6.626 \times 10^{-34}$ J s uchun V ning son qiymatini keltiring.

1.5 (2.0 ball) ϕ burchak $\phi = -90^\circ$ dan $\phi = 90^\circ$ gacha oshirilganda, OUT1 chiqish porti necha marta to'liq sikl (yuqori intensivlikdan past intensivlikka va yana yuqori intensivlikka qaytish)ni bajaradi?

Eksperimental ma'lumotlar (Experimental data)

Haqiqiy tajribadagi interferometr $a = 3.600$ cm va $\theta = 22.10^\circ$ bilan tavsiflangan va 19.00 to'liq sikl kuzatilgan.

1.6 (1.0 ball) Ushbu tajribada λ_0 qancha bo'lgan?

1.7 (1.0 ball) Agar xuddi shunday turdagi boshqa tajribada $\lambda_0 = 0.2000$ nm bo'lgan neytronlar yordamida 30.00 to'liq sikl kuzatilgan bo'lsa, A maydoni qancha bo'ladi?

Ko'rsatma (Hint): Agar $|\alpha x| \ll 1$ bo'lsa, $(1 + x)^\alpha$ ni $1 + \alpha x$ bilan almashtirish mumkin.

Javoblar varagi (Answer Script)

Geometriya (Geometry)

1.1 Maydon $A =$

1.2 Balandlik $H =$

Optik yol uzunligi (Optical path length)

1.3 $a, \theta, \phi, M, \lambda_0, g, h$ orqali:

$$\Delta N_{\text{opt}} =$$

1.4 A, V, λ_0, ϕ orqali:

$$\Delta N_{\text{opt}} =$$

V ning son qiymati:

$$V =$$

1.5 Sikllar soni:

$$\# \text{ of cycles} =$$

Eksperimental ma'lumotlar (Experimental data)

1.6 De Broyl to'lqin uzunligi:

$$\lambda_0 =$$

1.7 Maydon:

$$A =$$

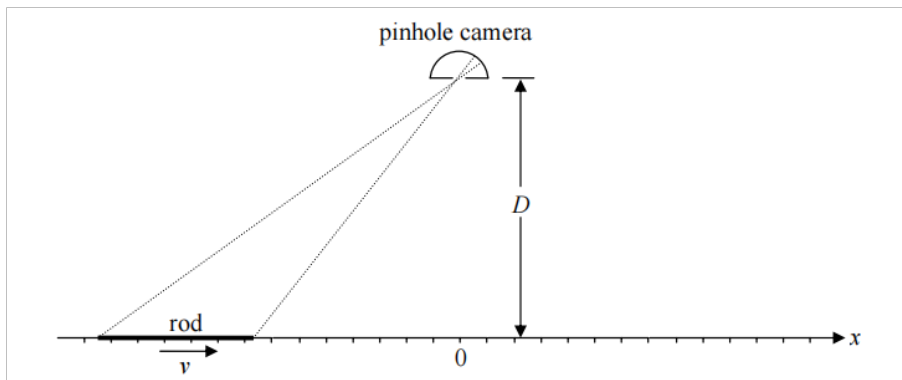
Nazariy masala 2: Harakatdagi sterjenni kuzatish

Fizik holat (Physical situation)

Pinhole kamerasi (pinhole camera), teshigi $x = 0$ da va x oqidan D masofada joylashgan, juda qisqa vaqtga teshikni ochib, sterjendan suratga oladi. x oqi boylab bir-biridan teng masofada belgilar mavjud bolib, ular yordamida sterjenning suratdagi korinma uzunligini (apparent length) aniqlash mumkin. Sterjen tinch holatda (at rest) bolganda uning uzunligi L ga teng. Ammo sterjen tinch holatda emas, balki x oqi boylab ozgarmas v tezlik bilan harakatlanmoqda.

Asosiy munosabatlar (Basic relations)

Pinhole kamerasi suratda sterjenning kichik bir bolagini $\bar{x}$ holatda korsatadi.



Pinhole kamera

geometriyasi.

- 2.1 (0.6 ball) Surat olingan paytda ushbu bolakning haqiqiy joylashuvi (actual position) x qancha? Javobingizni $\bar{x}$, D , L , v va yoruglik tezligi $c = 3.00 \times 10^8$ m/s orqali ifodalang. Natijangizni soddalashtirishda quyidagi kattaliklardan foydalaning:

$$\beta = \frac{v}{c}, \quad \gamma = \frac{1}{\sqrt{1 - \beta^2}}.$$

- 2.2 (0.9 ball) Teskari munosabatni toping, yani $\bar{x}$ ni x , D , L , v va c orqali ifodalang.

Eslatma: Haqiqiy joylashuv – kamera tinch turgan sanoq sistemasidagi joylashuv.

Sterjenning korinma uzunligi (Apparent length of the rod)

Pinhole kamerasi sterjen markazining haqiqiy joylashuvi x_0 bolgan paytda suratga oladi.

- 2.3 (1.5 ball) Berilgan ozgaruvchilar orqali ushbu suratdagi sterjenning korinma uzunligini aniqlang.
- 2.4 (1.5 ball) Javoblar varagida korinma uzunlik vaqt bilan qanday ozgarishini korsatuvchi katakchani belgilang.

Simmetrik rasm (Symmetric picture)

Bitta pinhole-kamera suratida sterjenning ikkala uchi pinhole teshigidan bir xil masofada korinadi.

- 2.5 (0.8 ball) Ushbu suratdagi sterjenning korinma uzunligini aniqlang.
- 2.6 (1.0 ball) Ushbu surat olingan paytda sterjen ortasining haqiqiy joylashuvi x_0 qanday?
- 2.7 (1.2 ball) Surat sterjen ortasining tasvirini (image) sterjen oldingi uchining tasviridan qancha masofada korsatadi?

Juda erta va juda kech suratlar (Very early and very late pictures)

Pinhole kamerasi juda erta bir surat oldi (sterjen juda uzoqda va yaqinlashayotgan paytda) va juda kech boshqa surat oldi (sterjen juda uzoqda va uzoqlashayotgan paytda). Suratlarining birida korinma uzunlik 1.00 m, ikkinchisida 3.00 m.

- 2.8 (0.5 ball) Javoblar varagida qaysi uzunlik qaysi suratda korilganini belgilang.
- 2.9 (1.0 ball) v tezlikni aniqlang.
- 2.10 (0.6 ball) Tinch holatdagi sterjen uzunligi L ni aniqlang.
- 2.11 (0.4 ball) Simmetrik suratdagi korinma uzunlikni toping.

Nazariy masala 3 (Theory Question 3)

Ushbu masala beshta mustaqil qismdan iborat. Har biri aniq javobni emas, balki faqat kattalik tartibini (order of magnitude) baholashni talab qiladi. Barcha javoblarni Javoblar varag'iga yozing.

Raqamli kamera (Digital Camera)

Kvadrat CCD chipi (CCD chip) chiziqli o'lchami $L = 35$ mm va $N_p = 5$ Mpix (1 Mpix = 10^6 piksel) bo'lgan raqamli kamerani ko'rib chiqaylik. Ushbu kameraning linzasi (lens) fokus masofasiga (focal length) ega $f = 38$ mm. Linzada ko'rinadigan taniqli raqamlar ketma-ketligi (2, 2.8, 4, 5.6, 8, 11, 16, 22) F-raqam (F-number) deb ataladi va u fokus masofasining linza diafragma (aperture) diametri D ga nisbati sifatida aniqlanadi: $F\# = f/D$.

- 3.1 (1.0 ball) Kameraning linza tomonidan cheklangan eng yaxshi mumkin bo'lgan fazoviy ajrata olish qobiliyatini (best possible spatial resolution) $\Delta x_{\min}$ ni chip (CCD) da toping. Natijangizni to'lqin uzunligi λ va F-raqam $F\#$ orqali ifodalang va $\lambda = 500$ nm uchun son qiymatini bering.
- 3.2 (0.5 ball) Ushbu optimal ajrata olish qobiliyatiga mos keladigan CCD chipdagi kerakli Mpix soni N ni toping.

- 3.3 (0.5 ball) Ba’zida fotografiyachilar kamerani eng kichik amaliy diafragmada ishlatishga harakat qilishadi. Faraz qilaylik, endi bizda chip o’lchami va fokus masofasi yuqorida berilganidek, $N_0 = 16$ Mpix bo’lgan kamera bor. Tasvir sifati optika tomonidan cheklanmasligi (image quality not limited by the optics) uchun $F\#$ ning qanday qiymati tanlanishi kerak?
- 3.4 (0.5 ball) Inson ko’zining taxminiy burchak ajrata olish qobiliyati (angular resolution) $\phi = 2$ yoy daqiqasi (arcmin) va odatdagi fotoprinter minimal 300 dpi (dots per inch) bosishini bilib, bosilgan sahifani ko’zingizdan qanday minimal masofada z ushlashingiz kerak, shunda alohida nuqtalarni (dots) ko’rmaysiz? Ma’lumot: 1 inch = 25.4 mm, 1 arcmin = 2.91×10^{-4} rad.

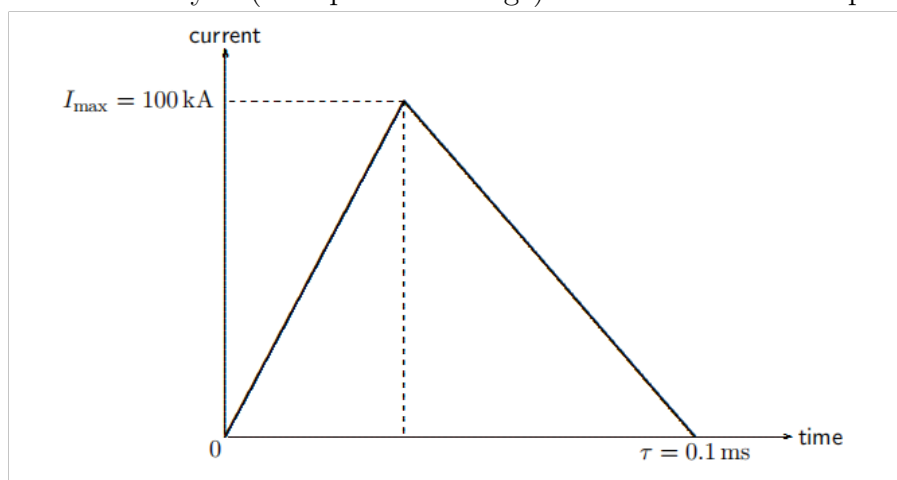
Qattiq qaynatilgan tuxum (Hard-boiled egg)

To’g’ridan-to’g’ri muzlatgichdan olingan $T_0 = 4^\circ\text{C}$ haroratli tuxum qaynayotgan suv (harorat $T_1 = 100^\circ\text{C}$) solingan idishga tashlanadi.

- 3.5 (0.5 ball) Tuxumning koagulyatsiyalanishi (qattiq bo’lishi) uchun qancha energiya U kerak?
- 3.6 (0.5 ball) Tuxum ichiga kiradigan issiqlik oqimi (heat flow) J qancha?
- 3.7 (0.5 ball) Tuxumga o’tkaziladigan issiqlik quvvati (heat power) P qancha?
- 3.8 (0.5 ball) Tuxumni qattiq qaynatilgan bo’lishi uchun qancha vaqt qaynatish kerak?

Chaqmoq (Lightning)

Chaqmoqning soddalashtirilgan modeli keltirilgan. Chaqmoq bulutlarda elektrostatik zaryad to’planishi natijasida yuzaga keladi. Natijada, bulutning pastki qismi odatda musbat, yuqori qismi manfiy zaryadlanadi va bulut ostidagi zamin manfiy zaryadlanadi. Tegishli elektr maydoni havoning uzilish kuchlanganligi (breakdown strength) qiymatidan oshganda, buzuvchi razryad (disruptive discharge) sodir bo’ladi: bu chaqmoqdir.



Quyidagi savollarga ushbu soddalashtirilgan tok vaqt funksiyasi egri chizig’i va ma’lumotlar yordamida javob bering:

- Bulut pastki qismi va zamin orasidagi masofa: $h = 1$ km;

- Nam havoning uzilish elektr maydoni (breakdown electric field): $E_0 = 300 \text{ kV/m}$;
- Yiliga Yerga tushadigan chaqmoqlarning umumiy soni: 32×10^6 ;
- Umumiy inson populyatsiyasi: 6.5×10^9 kishi.

3.9 (0.5 ball) Chaqmoq tomonidan chiqarilgan umumiy zaryad Q qancha?

3.10 (0.5 ball) Chaqmoq vaqtida bulut pastki qismi va zamin o'rtasida oqadigan o'rtacha tok I qancha?

3.11 (1.0 ball) Tasavvur qiling, bir yildagi barcha bo'ronlarning energiyasi yig'ilib, barcha odamlar o'rtasida teng taqsimlanadi. Sizning ulushingizga to'g'ri keladigan energiya bilan 100 Vt quvvatli lampochkani qancha vaqt davomida uzluksiz yorita olasiz?

Kapillyar tomirlar (Capillary Vessels)

Qonni suvga o'xshash massaviy zichlikka ega bo'lgan siqilmaydigan yopishqoq suyuqlik (incompressible viscous fluid) deb hisoblaymiz: zichlik μ va dinamik yopishqoqlik (dynamic viscosity) $\eta = 4.5 \text{ g m}^{-1}\text{s}^{-1}$. Qon tomirlarini radiusi r va uzunligi L bo'lgan aylana kesimli to'g'ri quvurlar (circular straight pipes) sifatida modellashtiramiz va qon oqimini Puazeyl qonuni (Poiseuille's law) bilan ifodalaymiz:

$$\Delta p = RD,$$

bu elektrdagi Om qonunining gidrodinamik analogidir. Bu yerda Δp – qon tomirining kirishi va chiqishi orasidagi bosim farqi, $D = S\nu$ – qon tomirining kesim yuzasi S orqali o'tadigan hajmli oqim (volume flow), ν – qon tezligi. Gidravlik qarshilik (hydraulic resistance) R quyidagicha beriladi:

$$R = \frac{8\eta L}{\pi r^4}.$$

Tizimli qon aylanishi (systemic blood circulation – chap qorinchadan o'ng bo'lmachaga oqadigan) uchun, dam olayotgan odamda qon oqimi $D \approx 100 \text{ cm}^3/\text{s}$. Barcha kapillyar tomirlar parallel ulangan va ularning har biri radiusi $r = 4 \mu\text{m}$, uzunligi $L = 1 \text{ mm}$ va $\Delta p = 1 \text{ kPa}$ bosim farqi ostida ishlaydi deb faraz qilib, quyidagi savollarga javob bering.

3.12 (1.0 ball) Inson tanasida nechta kapillyar tomir (capillary vessel) mavjud?

3.13 (0.5 ball) Kapillyar tomir orqali oqadigan qonning tezligi ν qancha?

Osmono'par bino (Skyscraper)

Balandligi 1000 m bo'lgan osmono'par binoning pastki qismida tashqi havo harorati $T_{\text{bot}} = 30^\circ\text{C}$. Yuqori qismdagi tashqi havo haroratini T_{top} baholash maqsadi. Bir oz yupqa havo qatlami (ideal azot gazi, adiabatik ko'rsatkich $\gamma = 7/5$) sekin-asta z balandlikka ko'tarilayotganini ko'rib chiqaylik, bu erda bosim pastroq. Ushbu qatlam adiabatik kengayadi deb faraz qiling, shunda uning harorati atrofdagi havo haroratiga teng bo'ladi.

3.14 (0.5 ball) Haroratning nisbiy o'zgarishi dT/T bosimning nisbiy o'zgarishi dp/p bilan qanday bog'liq?

3.15 (0.5 ball) Bosim farqini dp ni dz balandlik o'zgarishi orqali ifodalang.

3.16 (1.0 ball) Binoning yuqori qismida natijaviy harorat qanday?

Javoblar varag'i (Answer Script)

Raqamli kamera (Digital Camera)

3.1 Eng yaxshi fazoviy ajrata olish qobiliyati (Formula:) $\Delta x_{\min} =$

bu $\lambda = 500 \text{ nm}$ uchun (son qiymat:) $\Delta x_{\min} \approx$

3.2 Mpix soni $N \approx$

3.3 Eng yaxshi F-raqam qiymati $F\# \approx$

3.4 Minimal masofa $z \approx$

Country Code	Student Code	Question Number
		3

Qattiq qaynatilgan tuxum (Hard-boiled egg)

3.5 Kerakli energiya:

$$U = \underline{\hspace{2cm}}$$

3.6 Issiqlik oqimi:

$$J = \underline{\hspace{2cm}}$$

3.7 O'tkazilgan issiqlik quvvati:

$$P = \underline{\hspace{2cm}}$$

3.8 Tuxumni qattiq qaynatish uchun kerakli vaqt:

$$\tau = \underline{\hspace{2cm}}$$

Country Code	Student Code	Question Number
		3

Chaqmoq (Lightning)

3.9 Umumiy zaryad:

$$Q = \underline{\hspace{4cm}}$$

3.10 O'rtacha tok:

$$I = \underline{\hspace{4cm}}$$

3.11 Lampochka yonish vaqti:

$$t = \underline{\hspace{4cm}}$$

Kapillyar tomirlar (Capillary Vessels)

3.12 Inson tanasida kapillyar tomirlar soni:

$$N = \underline{\hspace{4cm}}$$

3.13 Qon tezligi:

$$v = \underline{\hspace{4cm}}$$

Country Code	Student Code	Question Number
		3

Osmono'par bino (Skyscraper)

3.14 Haroratning nisbiy o'zgarishi:

$$\frac{dT}{T} = \underline{\hspace{4cm}}$$

3.15 Bosim farqi:

$$dp = \underline{\hspace{4cm}}$$

3.16 Binoning yuqori qismidagi harorat:

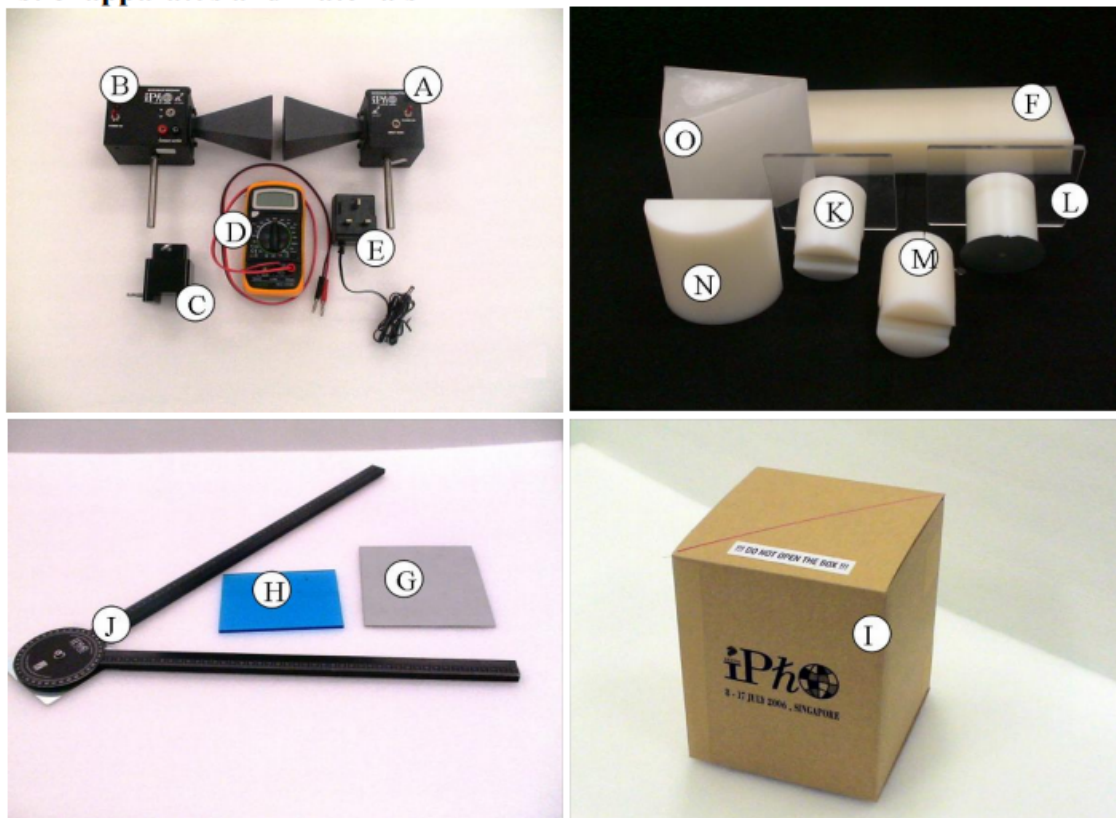
$$T_{\text{top}} = \underline{\hspace{4cm}}$$

37th International Physics Olympiad
Singapore
8 – 17 July 2006

Experimental Competition
Wed 12 July 2006

Asboblar va materiallar ro'yxati (List of apparatus and materials)

List of apparatus and materials



Belgi	Qism	Miqdor	Belgi	Qism	Miqdor
Ⓐ	Mikroto'lqin uzatgich (elektromagnit to'lqin tarqatuvchi uskuna)	1	Ⓘ	Qora qutidagi panjara strukturasini (kristall panjara modeli)	1
Ⓑ	Mikroto'lqin qabul qilgich (elektromagnit to'lqin tutuvchi uskuna)	1	⓵	Goniometr (burchaklarni o'lchovchi asbob)	1
Ⓒ	Uzatgich/qabul qilgich ushlagichi (qotiruvchi moslama)	2	⓶	Prizma ushlagichi (prizmani qotiruvchi moslama)	1
Ⓓ	Raqamli multimetr (kuchlanish, tok va qarshilik o'lchovchi asbob)	1	⓷	Aylanuvchi stol (aylanma harakatlanuvchi sirt)	1
Ⓔ	Uzatgich uchun o'zgarmas tok manbai (tok bilan ta'minlovchi uskuna)	1	⓸	Linza/qaytaruvchi ushlagichi (linza yoki ko'zguni qotiruvchi moslama)	1
Ⓕ	"Yupqa plyonka" namunasi sifatidagi plastinka (yupqa qatlamli namuna)	1	⓹	Yassi-silindrik linza (bir tomoni yassi, ikkinchi tomoni silindrik qavariq shisha)	1
Ⓖ	Qaytaruvchi (kumushsimon metall list - yorug'lik yoki to'lqinni qaytaruvchi sirt)	1	⓺	Mumli prizma (mumdan yasalgan prizma)	2
Ⓗ	Nurni ajratgich (ko'k rangli pleksiglas - nurni ikkiga bo'luvchi shisha)	1		Blu-Tack (yopishtiruvchi plastilin)	1 o'ram
	Shtangensirkul (alohida taqdim etiladi - yuqori aniqlikda uzunlik o'lchovchi asbob)			30 sm chizg'ich (alohida taqdim etiladi - uzunlik o'lchovchi asbob)	

Eksperimental masala (Experimental Competition)

Ehtiyot choralari (Caution):

- Mikroto'lqin uzatgichning (microwave transmitter) chiqish quvvati standart xavfsizlik darajalarida. Shunga qaramay, uzatgich yoqilgan holatda hech qachon mikroto'lqin gorniga yaqin masofadan to'g'ridan-to'g'ri qaramang.
- Panjara (lattice) solingan qutini () ochmang.
- Mum prizmalari () mort (3-qismda ishlatiladi).

Eslatma (Note):

- Mikroto'lqin qabul qilgichning (microwave receiver) chiqish TOKI (CURRENT) mikroto'lqinning AMPLITUDASIGA proporsional ekanligini ta'kidlash muhim.
- Har doim mikroto'lqin qabul qilgichning LO gain sozlamasidan foydalaning.
- Malumotlarni yig'ish vaqtida multimetr diapazonini o'zgartirmang.
- Interferensiyani minimallashtirish uchun ishlatilmaydigan komponentlarni tajriba joyidan uzoqroqqa qo'ying.
- Barcha chizmalaringizda komponentlarni ko'rsatish uchun har doim komponent belgilaridan (A, B, C, ...) foydalaning.

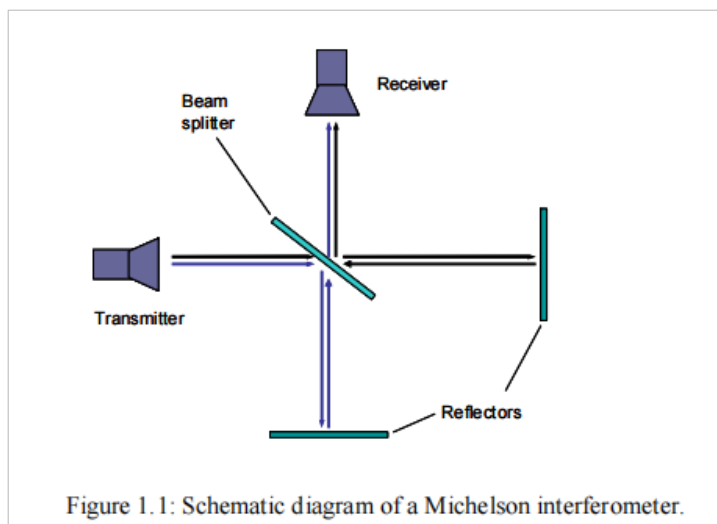


Raqamli multimetr (digital multimeter) ikkita sim bilan diagrammada ko'rsatilganidek ulanishi kerak. Ushbu tajribada siz "2m" tok sozlamasidan foydalanishingiz kerak.

1-qism: Mishelson interferometri (Michelson interferometer)

1.1. Kirish (Introduction)

Mishelson interferometrida (Michelson interferometer) nur ajratgich (beam splitter) kiruvchi elektromagnit (EM) to'lni ikki alohida yo'l bo'ylab yuboradi, so'ngra aks etishdan keyin tashkil etuvchi to'qlarni qayta birlashtiradi, shunda ular superpozitsiyalanib, interferensiyalar manzari (interference pattern) hosil qiladi. 1.1-rasm Mishelson interferometrining qurilmasini ko'rsatadi. Tushayotgan to'lin uzatgichdan qabul qilgichga ikki xil yo'l bo'ylab boradi. Bu ikki to'lin superpozitsiyalanadi va qabul qilgichda interferensiyalanadi. Qabul qilgichdagi signal kuchi ikki to'lin orasidagi fazalar farqiga (phase difference) bog'liq bo'lib, uni optik yo'l farqini (optical path difference) o'zgartirish orqali o'zgartirish mumkin.



1.1-rasm: Mishelson interferometrining sxematik diagrammasi.

1.2. Komponentlar ro'yxati (List of components)

1. A ushlagichi C bilan birga mikroto'lin uzatgich (Microwave transmitter A with holder C)
2. B ushlagichi C bilan birga mikroto'lin qabul qilgich (Microwave receiver B with holder C)
3. J Goniometr (Goniometer J)
4. 2 ta qaytargich: G qaytargich M ushlagichi bilan va yupqa plyonka F qaytargich vazifasida.
5. Nur ajratgich H aylanadigan stol J ushlagichi bilan (Beam splitter H with rotating table J acting as a holder)
6. D raqamli multimetr (Digital multimeter D)

1.3. Topshiriq: Mikroto'lqinning to'lqin uzunligini aniqlash [2 ball]

1.2-bo'limda keltirilgan eksperimental komponentlardan foydalanib, havodagi mikroto'lqinning to'lqin uzunligi λ ni aniqlash uchun Mishelson interferometri tajribasini o'rnatish. Ma'lumotlaringizni yozib oling va λ ni noaniqlik ≤ 0.02 sm bo'ladigan tarzda aniqlang.

Eslatma: "Yupqa plyonka" qisman o'tkazuvchan (partially transmissive), shuning uchun uning orqasida turmang yoki harakatlanmang, chunki bu natijalaringizga ta'sir qilishi mumkin.

2-qism: “Yupqa plyonka” interferensiyasi (Thin film interference)

2.1. Kirish (Introduction)

Dielektrik yupqa plyonkaga tushayotgan EM to'liq nuri 2.1-rasmda ko'rsatilganidek ikki nurga bo'linadi. A nurlari plyonkaning yuqori yuzasidan aks etadi, B nurlari esa plyonkaning pastki yuzasidan aks etadi. A va B nurlarining superpozitsiyasi "yupqa plyonka interferensiyasi" (thin film interference) deb ataladi.

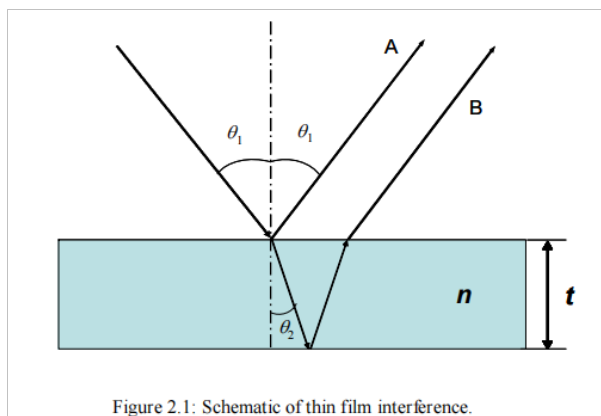


Figure 2.1: Schematic of thin film interference.

2.1-rasm: Yupqa plyonka interferensiyasining sxematik ko'rinishi.

A va B nurlarining optik yo'l uzunliklarining farqi konstruktiv (constructive) yoki destruktiv (destructive) interferensiyaga olib keladi. Natijaviy EM to'liq intensivligi I interferensiyalashuvchi ikki nurning yo'l farqiga bog'liq bo'lib, bu esa oz navbatida tushish burchagi θ_1 , nurlanish to'liq uzunligi λ hamda yupqa plyonkaning qalinligi t va sindirish ko'rsatkichiga n bog'liq. Shunday qilib, yupqa plyonkaning sindirish ko'rsatkichi n ni $I - \theta_1$ grafigidan, t va λ qiymatlaridan foydalanib aniqlash mumkin.

2.2. Komponentlar ro'yxati (List of components)

1. A mikroto'liq uzatgich (Microwave transmitter A) C ushlagichi bilan
2. B mikroto'liq qabul qilgich (Microwave receiver B) C ushlagichi bilan
3. N Plano-silindrik linza (Plano-cylindrical lens N) M ushlagichi bilan
4. J Goniometr (Goniometer J)
5. L Aylanadigan stol (Rotating table L)
6. D Raqamli multimetr (Digital multimeter D)
7. F Polimer plastinka – “yupqa plyonka” namunasi (Polymer slab acting as “thin film” sample F)
8. Noniusli shtangen-sirkul (Vernier caliper)

2.3. Topshiriqlar: Polimer plastinkaning sindirish ko'rsatkichini aniqlash [6 ball]

1. [1 ball] θ_1 , t , λ va n orqali konstruktiv va destruktiv interferensiyalar shartlari uchun ifodalarni keltirib chiqaring.
2. [3 ball] 2.2-bo'limda keltirilgan eksperimental komponentlardan foydalanib, qabul qilgich chiqishi S ni tushish burchagi θ_1 funksiyasi sifatida 40° dan 75° gacha bo'lgan oraliqda o'lchash uchun tajriba o'rnating. Eksperimental qurilmani chizing, tushish va qaytish burchaklarini hamda plyonkaning aylanadigan stoldagi holatini aniq ko'rsating. 2-sahifada berilgan yorliqlardan foydalanib barcha komponentlarni belgilang. Ma'lumotlaringizni jadvalga yozing. Qabul qilgich chiqishi S ni tushish burchagi θ_1 ga nisbatan grafigini chizing. Konstruktiv va destruktiv interferensiyalarga mos keladigan burchaklarni aniq aniqlang.
3. [2 ball] Havoning sindirish ko'rsatkichini 1.00 deb hisoblab, interferensiya tartibi m va polimer plastinkaning sindirish ko'rsatkichi n ni aniqlang. m va n qiymatlarini javoblar varag'iga yozing.

4. **[1.5 ball]** Natijalaringiz uchun xatolik tahlilini (error analysis) o'tkazing va n ning noaniqligini Δn baholang. Δn qiymatini javoblar varag'iga yozing.
5. **[0.5 ball]**

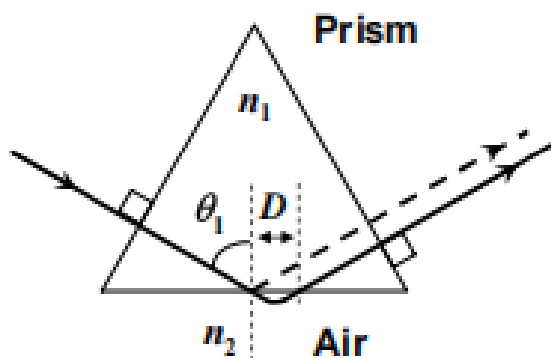
Eslatma (Note):

- Linza (lens) mikroto'lqin uzatgichning oldiga tekis yuzasi uzatgichga qaratilgan holda joylashtirilishi kerak, bu kvazi-parallel mikroto'lqin nurlarini olish uchun. Linzaning tekis yuzasi va uzatgich gorni diafragmasi orasidagi masofa 3 sm bo'lishi kerak.
- Eng yaxshi natijalar uchun uzatgich va qabul qilgich orasidagi masofani maksimal qiling.
- Uzatgich tomonidan chiqarilgan mikroto'lqinlarning tekis to'lqindan og'ishi (deviations) kuzatilgan manzarada qo'shimcha cho'qqilarga (extra peaks) olib kelishi mumkin. 40° dan 75° gacha bo'lgan belgilangan oraliqda interferensiya tufayli faqat bitta maksimum va bitta minimum mavjud.

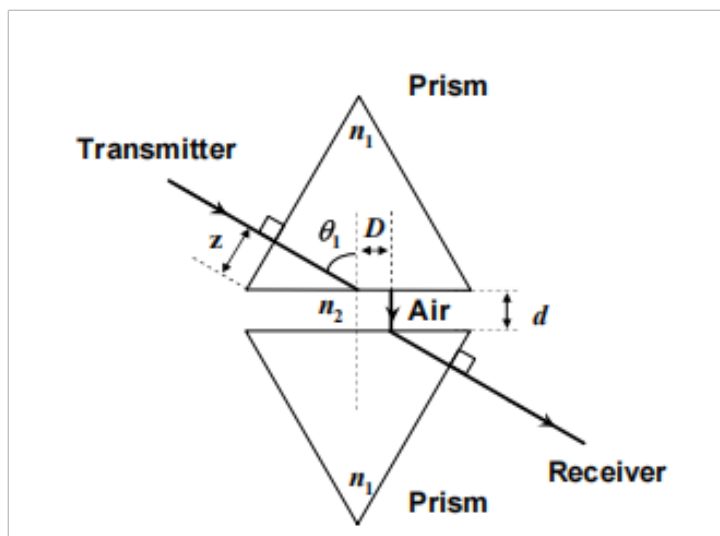
3-qism: Frustrated Total Internal Reflection (FTIR)

3.1. Kirish (Introduction)

To'liq ichki qaytish (total internal reflection – TIR) hodisasi tekis to'lqin optik zich muhitdan (optically dense medium) kam zich muhitga o'tganda sodir bo'lishi mumkin. Biroq, geometrik optika tomonidan bashorat qilingan chegaradagi TIR o'rniga, kiruvchi to'lqin haqiqatda kam zich muhitga kirib boradi va chegaraga parallel ravishda bir masofa bosib o'tgandan so'ng zichroq muhitga qaytib sochiladi (3.1-rasmga qarang). Ushbu effekt qaytgan nurning siljishi D bilan tavsiflanishi mumkin, bu Goos–Hänchen siljishi (Goos–Hänchen shift) deb ataladi.



3.1-rasm: Prizmada to'liq ichki qaytishga uchrayotgan EM to'lqin. Havodagi sirtga parallel bo'lgan D siljishi Goos–Hänchen siljishini ifodalaydi.



3.2-rasm: Prizmalar va d masofali havo bo'shlig'ini ko'rsatuvchi eksperimental qurilma chizmasi. D siljishi – Goos–Hänchen siljishi. z – prizma uchidan uzatgichning markaziy o'qigacha bo'lgan masofa.

Agar birinchi muhit bilan bir xil sindirish ko'rsatkichiga ega bo'lgan boshqa muhit (n_1) birinchi muhitga 3.2-rasmga ko'rsatilganidek kichik d masofada joylashtirilsa, EM to'lqinning ikkinchi muhit orqali tunnel o'tishi (tunneling) sodir bo'ladi. Ushbu qiziqarli

hodisa to'siqli to'liq ichki qaytish (frustrated total internal reflection – FTIR) deb ataladi. O'tayotgan to'lqinning intensivligi I_t masofa d bilan eksponensial ravishda kamayadi:

$$I_t = I_0 \exp(-2\gamma d) \quad (3.1)$$

bu yerda I_0 – tushayotgan to'lqinning intensivligi va γ :

$$\gamma = \frac{2\pi}{\lambda} \sqrt{\frac{n_1^2}{n_2^2} \sin^2 \theta_1 - 1} \quad (3.2)$$

bu yerda λ – 2-muhitdagi EM to'lqin uzunligi, n_2 – 2-muhitning sindirish ko'rsatkichi (2-muhit – havo, sindirish ko'rsatkichi 1.00 deb faraz qiling).

3.2. Komponentlar ro'yxati (List of components)

1. Mikroto'lqin uzatgich A, ushlagich C bilan (Microwave transmitter A with holder C)
2. Mikroto'lqin qabul qilgich B, ushlagich C bilan (Microwave receiver B with holder C)
3. Plano-silindrik linza N, ushlagich M bilan (Plano-cylindrical lens N with holder M)
4. 2 ta teng tomonli mum prizmasi O, ushlagich K va aylanadigan stol L (2 equilateral wax prisms O with holder K and rotating table L acting as a holder)
5. Raqamli multimetr D (Digital multimeter D)
6. Goniometr J (Goniometer J)
7. Chizg'ich (Ruler)

3.3. Tajriba tavsifi (Description of the Experiment)

3.2-bo'limda tavsiflangan komponentlar ro'yxatidan foydalanib, FTIRda I_t intensivlikning havo bo'shlig'i d ga bog'liqligini o'rganish uchun tajriba o'rnating. Barqaror natijalar uchun quyidagilarga e'tibor bering:

- Ushbu tajriba uchun goniometrning bir qo'lidan foydalaning.
- Prizma sirtlarini bir-biriga parallel bo'lishi uchun ehtiyotkorlik bilan tanlang.
- Linzaning egri yuzasi markazidan prizma sirtigacha bo'lgan masofa 2 sm bo'lishi kerak.
- Detektorni uning gorni prizma yuziga tegib turadigan qilib joylashtiring.
- Har bir d qiymati uchun maksimal signalni olish uchun mikroto'lqin qabul qilgichning holatini prizma yuzasi bo'ylab sozlang.
- Raqamli multimetrning 2 mA diapazonida ekanligiga ishonch hosil qiling. Ma'lumotlarni $d = 0.6$ sm dan boshlab to'plang. Multimetr ko'rsatkichi 0.20 mA dan pastga tushganda o'lchovlarni to'xtating.

3.4. Topshiriqlar: Prizma materialining sindirish ko'rsatkichini aniqlash [6 ball]

1-topshiriq [1 ball] Yakuniy eksperimental qurilmani chizing va 2-sahifada berilgan yorliqlardan foydalanib barcha komponentlarni belgilang. Chizmangizda z masofasining qiymatini (3.2-rasmga qarang – prizma uchidan uzatgichning markaziy o'qigacha bo'lgan masofa) yozib qo'ying.

2-topshiriq [2.1 ball] Tajribangizni o'tkazing va ma'lumotlaringizni jadvalga yozing. Ushbu topshiriqni ikki marta bajaring.

3-topshiriq [2.9 ball]

- (a) Tegishli grafiklarni chizish orqali prizmaning sindirish ko'rsatkichi n_1 ni xatolik tahlili bilan aniqlang.
- (b) Prizmaning sindirish ko'rsatkichi n_1 va uning noaniqligi Δn_1 ni berilgan javoblar varag'iga yozing.

4-qism: Metall-rod panjarasining mikroto'lqin diffraksiyasi: Bragg qaytishi (Bragg reflection)

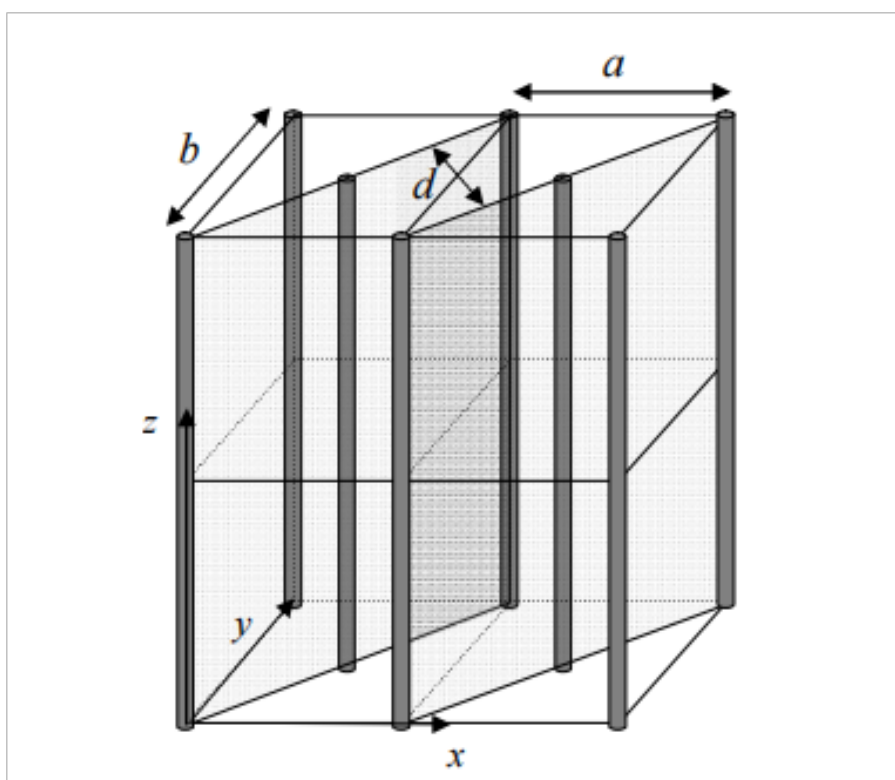
4.1. Kirish (Introduction)

Bragg qonuni (Bragg's Law) Haqiqiy kristallning panjara tuzilishini (lattice structure) Bragg qonuni yordamida tekshirish mumkin:

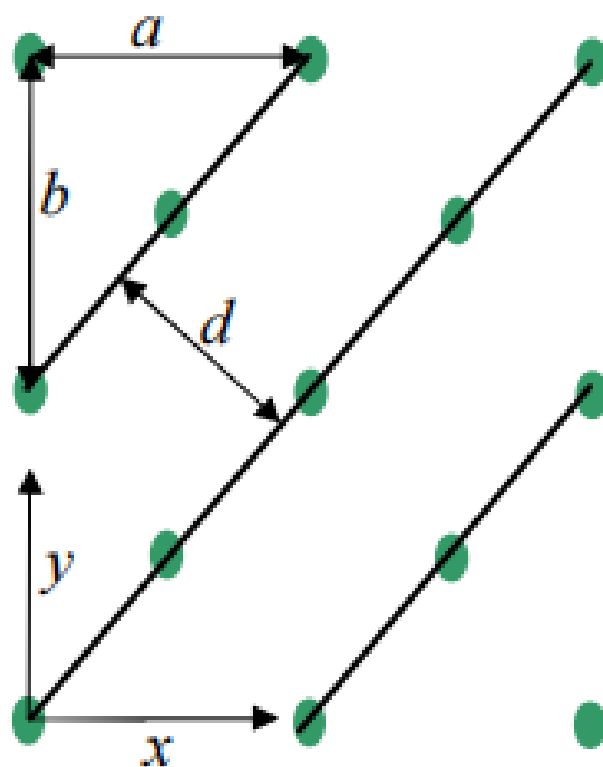
$$2d \sin \theta = m\lambda \quad (4.1)$$

bu yerda d – rentgen nurlarini "qaytaruvchi" parallel kristall tekisliklari to'plami orasidagi masofa; m – diffraksiya tartibi (order of diffraction); θ – tushayotgan rentgen nurlari va kristall tekisliklari orasidagi burchak. Bragg qonuni odatda Bragg qaytishi (Bragg reflection) yoki rentgen nurlari diffraksiyasi (X-ray diffraction) deb ham ataladi.

Metall-rod panjarasi (Metal-rod lattice) Rentgen nurlarining to'lqin uzunligi kristallning panjara konstantasiga (lattice constant) o'xshash bo'lganligi sababli, an'anaviy Bragg diffraksiyasi tajribasi rentgen nurlari yordamida amalga oshiriladi. Biroq, mikroto'lqinlar uchun diffraksiya ancha katta panjara konstantasiga ega bo'lgan panjara tuzilmalarida sodir bo'ladi, bu esa o'lchagich bilan oson o'lchanishi mumkin.



4.1-rasm: Panjara konstantalari a va b , hamda tekisliklar orasidagi masofa d bo'lgan metall-rod panjarasi.



4.2-rasm:

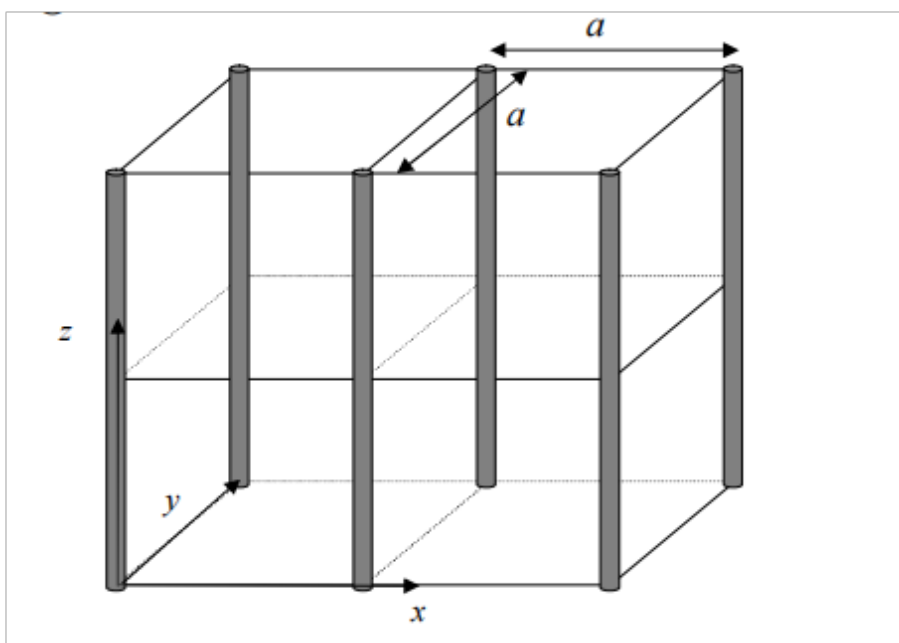
4.1-rasmdagi metall-rod panjarasining ustki ko'rinishi (masshtabga rioya qilinmagan).
Chiziqlar panjaraning diagonal tekisliklarini bildiradi.

4-qism: Metall-rod panjarasining mikroto'lqin diffraksiyasi: Bragg qaytishi (davomi)

Ushbu tajribada Bragg qonuni metall rodlardan yasalgan panjaraning (lattice) panjara konstantasini (lattice constant) o'lchash uchun ishlatiladi. Bunday metall-rod panjarasiga misol 4.1-rasmda ko'rsatilgan, bu yerda metall rodlar qalin vertikal chiziqlar sifatida ko'rsatilgan. xy -tekisligining diagonal yo'nalishi bo'ylab panjara tekisliklari (lattice planes) soyalangan tekisliklar sifatida ko'rsatilgan. 4.2-rasm metall-rod panjarasining ustki ko'rinishini (z o'qi bo'ylab pastga qarab) ko'rsatadi, bu nuqtalar rodلarni, chiziqlar esa diagonal panjara tekisliklarini bildiradi.

4.2. Komponentlar ro'yxati (List of components)

1. Mikroto'lqin uzatgich A, ushlagich C bilan (Microwave transmitter A with holder C)
2. Mikroto'lqin qabul qilgich B, ushlagich C bilan (Microwave receiver B with holder C)
3. Plano-silindrik linza N, ushlagich M bilan (Plano-cylindrical lens N with holder M)
4. Metall-rod panjarasi I solingan yopiq quti (Sealed box containing a metal-rod lattice I)
5. Aylanadigan stol L (Rotating table L)
6. Raqamli multimetr D (Digital multimeter D)
7. Goniometr J (Goniometer J)



4.3-rasm: Oddiy kvadrat panjara (simple square lattice).

Ushbu tajribada sizga 4.3-rasmda ko'rsatilganidek, metall rodlardan yasalgan oddiy kvadrat panjara berilgan. Panjara qutiga muhrlangan. Tajribadan panjaraning panjara konstantasi a ni chiqarishingiz so'raladi. QUTINI OCHMANG. Agar tajribadan keyin muhr buzilgan holda topilsa, eksperimental natijalar uchun ball berilmaydi.

4.3. Topshiriqlar: Berilgan oddiy kvadrat panjaraning panjara konstantasini aniqlash [6 ball]

1-topshiriq [1 ball] 4.3-rasmda ko'rsatilgan oddiy kvadrat panjaraning ustki ko'rinish diagrammasini chizing. Diagrammada berilgan panjaraning panjara konstantasi a va diagonal tekisliklarning tekisliklararo masofasini (interplanar spacing) d belgilang. Ushbu diagramma yordamida Bragg qonunini (Bragg's Law) keltirib chiqaring.

2-topshiriq [1.5 ball] Bragg qonuni va berilgan asboblardan foydalanib, panjaraning panjara konstantasi a ni aniqlash uchun Bragg diffraksiyasi tajribasini o'tkazishni loyihalashtiring.

- (a) Eksperimental qurilmani chizing. 2-sahifadagi yorliqlardan foydalanib barcha komponentlarni belgilang va uzatgich o'qi bilan panjara tekisliklari orasidagi burchak θ hamda uzatgich o'qi va qabul qilgich o'qi orasidagi burchak ζ ni aniq ko'rsating. Tajribangizda qutidagi qizil chiziqli bilan ko'rsatilgan diagonal tekisliklarda diffraksiyani o'lchang.

3-topshiriq [1.4 ball] $20^\circ \leq \theta \leq 50^\circ$ oralig'ida diffraksiya tajribasini o'tkazing. Ushbu oraliqda siz faqat birinchi tartibli diffraksiyani (first order diffraction) kuzatasiz. Javoblar varag'ida natijalaringizni jadvalga yozing va θ va ζ ni qayd eting.

4-topshiriq [1.3 ball] Diffraktsiyalangan to'lqin intensivligiga proporsional bo'lgan kattalikni θ ga nisbatan grafigini chizing.

5-topshiriq [0.8 ball] Grafikdan foydalanib, panjara konstantasi a ni aniqlang va eksperimental xatolikni baholang.

Eslatma (Note):

1. Eng yaxshi natijalar uchun uzatgich tajriba davomida qo'zg'almas holda qolishi kerak. Uzatgich va panjara orasidagi masofa, shuningdek panjara va qabul qilgich orasidagi masofa taxminan 50 sm bo'lishi kerak.
2. Ushbu tajribada faqat diagonal tekisliklardan foydalaning. Agar boshqa tekisliklarni ishlatishga harakat qilsangiz, natijangiz to'g'ri bo'lmaydi.
3. Panjara qutisining qizil diagonal chiziqli tomoni tepada bo'lishi kerak.
4. Diffraksiya cho'qqisining o'rnini yaxshiroq aniqlik bilan topish uchun cho'qqi atrofida bir nechta ma'lumot nuqtalaridan foydalaning.

1-masala yechimlari (Solutions to Theory Question 1)

Geometriya (Geometry)

Olmos shaklidagi maydonning har bir tomonining uzunligi $L = \frac{a}{\cos \theta}$ va parallel tomonlar orasidagi masofa $D = \frac{a}{\cos \theta} \sin(2\theta) = 2a \sin \theta$. Maydon $A = LD$:

$$A = 2a^2 \tan \theta.$$

OUT1 chiqish portini ϕ burchakka og'irish natijasida ko'tarilgan balandlik $H = D \sin \phi$:

$$H = 2a \sin \theta \sin \phi.$$

Optik yo'l uzunligi (Optical path length)

Faqat IN va OUT1 uchun ikkita parallel chiziq muhim, har birining uzunligi L . IN tomonida de Broyl to'lqin uzunligi λ_0 , OUT1 tomonida λ_1 bo'lsa:

$$\Delta N_{\text{opt}} = \frac{L}{\lambda_0} - \frac{L}{\lambda_1} = \frac{a}{\lambda_0 \cos \theta} \left(1 - \frac{\lambda_0}{\lambda_1} \right).$$

Impuls (momentum) mos ravishda h/λ_0 yoki h/λ_1 ga teng. Energiyaning saqlanish qonuni:

$$\frac{1}{2M} \left(\frac{h}{\lambda_0} \right)^2 = \frac{1}{2M} \left(\frac{h}{\lambda_1} \right)^2 + MgH,$$

bundan:

$$\frac{\lambda_0}{\lambda_1} = \sqrt{1 - 2 \frac{gM^2}{h^2} \lambda_0^2 H}.$$

$(gM^2/h^2)\lambda_0^2 H$ miqdori taxminan 10^{-7} tartibida ekanligini hisobga olib, soddalashtirsak:

$$\frac{\lambda_0}{\lambda_1} = 1 - \frac{gM^2}{h^2} \lambda_0^2 H,$$

va natijada:

$$\Delta N_{\text{opt}} = \frac{a}{\lambda_0 \cos \theta} \frac{gM^2}{h^2} \lambda_0^2 H$$

yoki

$$\Delta N_{\text{opt}} = 2 \frac{gM^2}{h^2} a^2 \lambda_0 \tan \theta \sin \phi. \quad (1.3)$$

Buni yanada ixchamroq yozish mumkin:

$$\Delta N_{\text{opt}} = \frac{\lambda_0 A}{V} \sin \phi, \quad (1.4)$$

bu yerda

$$V = \frac{h^2}{gM^2} = 0.1597 \times 10^{-13} \text{ m}^3 = 0.1597 \text{ nm} \cdot \text{cm}^2 \quad (1.4)$$

hajm parametrining (volume parameter) son qiymati.

Konstruktiv interferensiya (OUT1 da yuqori intensivlik) $\Delta N_{\text{opt}} = 0, \pm 1, \pm 2, \dots$ bo'lganda, destruktiv interferensiya (OUT1 da past intensivlik) esa $\Delta N_{\text{opt}} = \pm \frac{1}{2}, \pm \frac{3}{2}, \pm \frac{5}{2}, \dots$ bo'lganda sodir bo'ladi. ϕ ni $\phi = -90^\circ$ dan $\phi = 90^\circ$ gacha o'zgartirish:

$$\Delta N_{\text{opt}} \bigg|_{\phi=-90^\circ}^{90^\circ} = \frac{2\lambda_0 A}{V},$$

shuning uchun sikllar soni:

$$\# \text{ of cycles} = \frac{2\lambda_0 A}{V}. \quad (1.5)$$

Eksperimental ma'lumotlar (Experimental data)

$a = 3.6 \text{ cm}$ va $\theta = 22.1^\circ$ uchun $A = 2a^2 \tan \theta = 2 \cdot (3.6)^2 \cdot \tan 22.1^\circ = 10.53 \text{ cm}^2$. Shunday qilib,

$$\lambda_0 = \frac{19 \times 0.1597}{2 \times 10.53} \text{ nm} = 0.1441 \text{ nm}. \quad (1.6)$$

$\lambda_0 = 0.2 \text{ nm}$ bo'lgan neytronlar bilan 30 to'liq sikl kuzatilganda maydon:

$$A = \frac{30 \times 0.1597}{2 \times 0.2} \text{ cm}^2 = 11.98 \text{ cm}^2. \quad (1.7)$$

2-masala yechimlari (Solutions to Theory Question 2)

Asosiy munosabatlar (Basic relations)

Suratda $\tilde{x}$ ko'rinadi, agar yorug'lik shu joydan suratga olish vaqtidan yorug'likning T vaqt oldin chiqarilgan bo'lsa, T quyidagicha beriladi:

$$T = \sqrt{D^2 + \tilde{x}^2}/c.$$

T vaqt oralig'ida sterjenning tegishli bo'lagi vT masofaga harakatlanadi, shuning uchun suratga olish vaqtidagi uning haqiqiy joylashuvi x :

$$x = \tilde{x} + \beta \sqrt{D^2 + \tilde{x}^2}. \quad (2.1)$$

$\tilde{x}$ uchun yechib topamiz:

$$\tilde{x} = \gamma^2 x - \beta \gamma \sqrt{D^2 + (\gamma x)^2}. \quad (2.2)$$

Sterjenning korinma uzunligi (Apparent length of the rod)

Lorents qisqarishi (Lorentz contraction) tufayli harakatdagi sterjenning haqiqiy uzunligi L/γ ga teng, shuning uchun sterjen ikki uchining haqiqiy joylashuvlari:

$$x_{\pm} = x_0 \pm \frac{L}{2\gamma} \quad \text{uchun} \quad \left\{ \begin{array}{l} \text{oldingi uchi (front end)} \\ \text{orqadagi uchi (rear end)} \end{array} \right\}$$

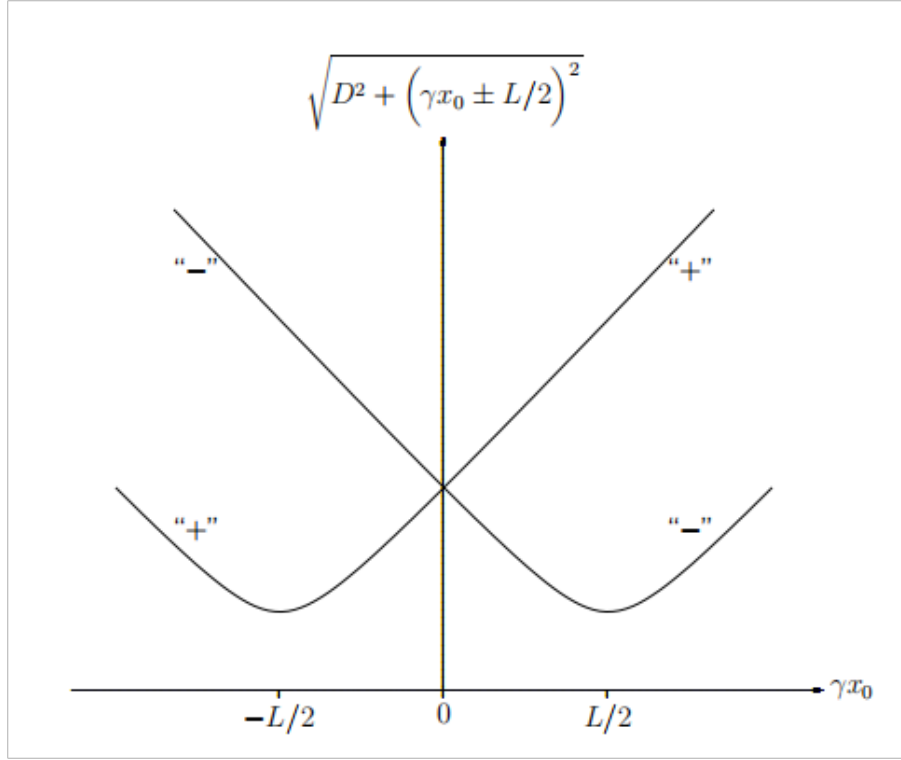
Pinhole kamerasi tomonidan olingan suratda sterjen uchlarining tasvirlari quyidagi joylarda ko'rinadi:

$$\tilde{x}_{\pm} = \gamma \left(\gamma x_0 \pm \frac{L}{2} \right) - \beta \gamma \sqrt{D^2 + \left(\gamma x_0 \pm \frac{L}{2} \right)^2}.$$

Korinma uzunlik $\tilde{L}(x_0) = \tilde{x}_+ - \tilde{x}_-$ shuning uchun:

$$\tilde{L}(x_0) = \gamma L + \beta \gamma \sqrt{D^2 + \left(\gamma x_0 - \frac{L}{2}\right)^2} - \beta \gamma \sqrt{D^2 + \left(\gamma x_0 + \frac{L}{2}\right)^2}. \quad (2.3)$$

Sterjen ozgarmas v tezlik bilan harakatlanayotgani uchun $\frac{dx_0}{dt} = v$ va shuning uchun savol $\tilde{L}(x_0)$ x_0 ortishi bilan ortadimi yoki kamayadimi. Ikki kvadrat ildiz hadini chizamiz:



Ikki kvadrat ildiz hadining grafigi.

$\tilde{L}(x_0)$ ifodasida “-” va “+” ishorali kvadrat ildizlarning ayirmasi paydo bo’ladi va bu ayirma x_0 ortishi bilan aniq kamayadi.

2.4

Korinma uzunlik doim kamayadi (apparent length decreases all the time). Javob: [x]
doim kamayadi.

Simmetrik rasm (Symmetric picture)

Simmetriya sabablariga ko'ra, simmetrik suratdagi korinma uzunlik harakatdagi sterjenning haqiqiy uzunligiga teng, chunki ikki uchidan chiqqan yorug'lik bir vaqtda pinhole teshigiga yetib kelishi uchun bir vaqtda chiqarilgan bo'lishi kerak, ya'ni:

$$\tilde{L} = \frac{L}{\gamma}. \quad (2.5)$$

Korinma uchlarning joylashuvlari $\tilde{x}_- = -\tilde{x}_+$ shartini qanoatlantiradi, yoki:

$$0 = \tilde{x}_+ + \tilde{x}_- = 2\gamma^2 x_0 - \beta\gamma\sqrt{D^2 + \left(\gamma x_0 + \frac{L}{2}\right)^2} - \beta\gamma\sqrt{D^2 + \left(\gamma x_0 - \frac{L}{2}\right)^2}. \quad (2.6)$$

Bunga qo'shimcha ravishda:

$$\frac{L}{\gamma} = \tilde{x}_+ - \tilde{x}_- = \gamma L - \beta\gamma\sqrt{D^2 + \left(\gamma x_0 + \frac{L}{2}\right)^2} + \beta\gamma\sqrt{D^2 + \left(\gamma x_0 - \frac{L}{2}\right)^2}$$

bu bizga quyidagini beradi:

$$\sqrt{D^2 + \left(\gamma x_0 \pm \frac{L}{2}\right)^2} = \frac{2\gamma^2 x_0 \pm (\gamma L - L/\gamma)}{2\beta\gamma} = \frac{\gamma x_0}{\beta} \pm \frac{\beta L}{2}.$$

Yuqori va pastki ishoralar bilan olingan ikkala versiya ham x_0 uchun bir xil javobni beradi, ya'ni:

$$x_0 = \beta\sqrt{D^2 + \left(\frac{L}{2\gamma}\right)^2}. \quad (2.7)$$

Simmetrik suratda sterjen ortasining tasviri, shuning uchun, quyidagi joyda joylashgan:

$$\begin{aligned} \tilde{x}_0 &= \gamma^2 x_0 - \beta\gamma\sqrt{D^2 + (\gamma x_0)^2} \\ &= \beta\gamma\left(\sqrt{(\gamma D)^2 + \left(\frac{L}{2}\right)^2} - \sqrt{(\gamma D)^2 + \left(\frac{\beta L}{2}\right)^2}\right). \end{aligned} \quad (2.8)$$

Bu tasvir oldingi uchi tasviridan $\ell = \tilde{x}_+ - \tilde{x}_0 = \frac{L}{2\gamma} - \tilde{x}_0$ masofada joylashgan, ya'ni:

$$\ell = \frac{L}{2\gamma} - \beta\gamma\sqrt{(\gamma D)^2 + \left(\frac{L}{2}\right)^2} + \beta\gamma\sqrt{(\gamma D)^2 + \left(\frac{\beta L}{2}\right)^2}$$

yoki

$$\ell = \frac{L}{2\gamma} \left[1 - \frac{\frac{\beta L}{2}}{\sqrt{(\gamma D)^2 + \left(\frac{L}{2}\right)^2} + \sqrt{(\gamma D)^2 + \left(\frac{\beta L}{2}\right)^2}} \right]. \quad (2.9)$$

Juda erta va juda kech suratlar (Very early and very late pictures)

Juda erta vaqtda x_0 juda katta manfiy qiymatga ega, shuning uchun juda erta suratdagi korinma uzunlik:

$$\tilde{L}_{\text{early}} = \tilde{L}(x_0 \rightarrow -\infty) = (1 + \beta)\gamma L = \sqrt{\frac{1 + \beta}{1 - \beta}} L. \quad (2.10)$$

Xuddi shunday, juda kech vaqtda x_0 juda katta musbat qiymatga ega, shuning uchun juda kech suratdagi korinma uzunlik:

$$\tilde{L}_{\text{late}} = \tilde{L}(x_0 \rightarrow \infty) = (1 - \beta)\gamma L = \sqrt{\frac{1 - \beta}{1 + \beta}} L. \quad (2.11)$$

Bundan $\tilde{L}_{\text{early}} > \tilde{L}_{\text{late}}$ ekanligi kelib chiqadi, ya'ni:

2.8

Korinma uzunlik erta suratda 3 m, kech suratda 1 m. Javob: Erta suratda 3 m, kech suratda 1 m.

Bundan tashqari,

$$\beta = \frac{\tilde{L}_{\text{early}} - \tilde{L}_{\text{late}}}{\tilde{L}_{\text{early}} + \tilde{L}_{\text{late}}},$$

shuning uchun $\beta = \frac{1}{2}$ va tezlik:

$$v = \frac{c}{2}. \quad (2.9)$$

Bundan $\gamma = \frac{\tilde{L}_{\text{early}} + \tilde{L}_{\text{late}}}{2\sqrt{\tilde{L}_{\text{early}}\tilde{L}_{\text{late}}}} = \frac{2}{\sqrt{3}} = 1.1547$. Quyidagi bilan birgalikda:

$$L = \sqrt{\tilde{L}_{\text{early}}\tilde{L}_{\text{late}}} = 1.73 \text{ m}, \quad (2.10)$$

bu simmetrik suratdagi uzunlikni beradi:

$$\tilde{L} = \frac{2\tilde{L}_{\text{early}}\tilde{L}_{\text{late}}}{\tilde{L}_{\text{early}} + \tilde{L}_{\text{late}}} = 1.50 \text{ m}. \quad (2.11)$$

3-masala yechimlari (Solutions to Theory Question 3)

Raqamli kamera (Digital Camera)

Kameraning ajrata olish qobiliyatini (resolution) cheklaydigan ikkita omil: diafragmadagi diffraktsiya (diffraction) va piksel o'lchami. Diffraktsiya uchun ajrata olishning burchak chegarasi (angular resolution) θ_R yorug'lik to'lqin uzunligi λ ning kamera diafragma diametri D ga nisbati:

$$\theta_R = 1.22 \frac{\lambda}{D},$$

bu yerda standart 1.22 koeffitsiyenti diafragmaning dumaloq shaklini aks ettiradi. Suratga olishda ob'ekt odatda fotografdan yetarlicha uzoqda bo'lgani uchun tasvir kameraning fokal tekisligida (focal plane) hosil bo'ladi va CCD chipi shu joyga joylashtirilishi kerak. Rayleigh diffraktsiya mezoni (Rayleigh diffraction criterion) shuni ko'rsatadiki, ikkita tasvir nuqtasini ajratish mumkin, agar ular bir-biridan

$$\Delta x = f\theta_R = 1.22\lambda F_{\#}$$

masofadan ko'proq uzoqlashgan bo'lsa (bu yerda $F_{\#} = f/D$ – F-raqam). Eng katta mumkin bo'lgan diafragma (yoki eng kichik $F_{\#} = 2$) va $\lambda = 500 \text{ nm}$ (kun yorug'ligining tipik to'lqin uzunligi) deb hisoblasak:

$$\Delta x = 1.22 \cdot 500 \cdot 10^{-9} \cdot 2 \approx 1.22 \mu\text{m}.$$

Raqamli ajrata olish qobiliyati (digital resolution) ikki qo'shni piksel markazlari orasidagi masofa ℓ bilan beriladi. 5 Mpx kameramiz uchun bu masofa taxminan:

$$\ell = \frac{L}{\sqrt{N_p}} = \frac{35 \times 10^{-3}}{\sqrt{5 \times 10^6}} \approx 15.65 \mu\text{m}.$$

Ideal holda optik va raqamli ajrata olish qobiliyatini moslashtirish kerak, shunda hech bir tomon ortiqcha belgilanmagan bo'ladi. Berilgan optik ajrata olish qobiliyatini raqamli ajrata olish qobiliyati ifodasiga qo'yib:

$$N = \left(\frac{L}{\Delta x} \right)^2 \approx 823 \text{ Mpix.} \quad (3.2)$$

Endi noma'lum optimal diafragmani topish uchun $\ell \geq \Delta x$ bo'lishi kerakligini eslaymiz, ya'ni $F_{\#} \leq F_0$ bilan:

$$F_0 = \frac{L}{1.22\lambda\sqrt{N_0}} = 2\sqrt{\frac{N}{N_0}} = 14.34. \quad (3.3)$$

Raqamli kamera (davomi)

Ushbu F_{eff} qiymati mavjud emasligi sababli, biz eng yaqin yuqori optik ajrata olish qobiliyatiga ega qiymatni tanlaymiz:

$$F_0 = 11. \quad (3.3)$$

Rasmga masofada z dan qaralganda, ikkita qo'shni nuqta orasidagi kichik burchak $\phi = \ell/z$, bu yerda ℓ – qo'shni nuqtalar orasidagi masofa. Shunga ko'ra,

$$z = \frac{\ell}{\phi} = \frac{2.54 \times 10^{-2}/300}{5.82 \times 10^{-4}} = 14.55 \text{ sm} \approx 15 \text{ sm}. \quad (3.4)$$

Qattiq qaynatilgan tuxum (Hard-boiled egg)

Tuxumning hammasi koagulyatsiya haroratiga (coagulation temperature) yetishi kerak. Bu harorat o'sishi:

$$\Delta T = T_c - T_0 = 65^\circ\text{C} - 4^\circ\text{C} = 61^\circ\text{C}.$$

Shunday qilib, tuxumning butunlay koagulyatsiyalanishi uchun unga kiritilishi kerak bo'lgan minimal energiya miqdori $U = \mu V C \Delta T$ bilan beriladi, bu yerda $V = \frac{4}{3}\pi R^3$ – tuxum hajmi. Shunday qilib:

$$U = \mu \cdot \frac{4\pi R^3}{3} \cdot C \cdot (T_c - T_0) = 1000 \cdot \frac{4\pi(0.025)^3}{3} \cdot 4200 \cdot 61 \approx 16768 \text{ J}. \quad (3.5)$$

Issiqlik oqimi (heat flow) uchun soddalashtirilgan tenglama vaqt birligida tuxum yuzasi orqali unga qancha energiya o'tganini hisoblash imkonini beradi. Vaqtning taxminiy qiymatini olish uchun tuxum markazining boshlang'ich harorati $T = 4^\circ\text{C}$ deb faraz qilaylik. Tipik uzunlik shkalasi (length scale) $\Delta r = R$ va u bilan bog'liq harorat farqi $\Delta T = T_1 - T_0$, bu yerda $T_1 = 100^\circ\text{C}$ (qaynayotgan suv). Shunday qilib:

$$J = \kappa \frac{T_1 - T_0}{R} = 0.64 \cdot \frac{96}{0.025} = 2458 \text{ W/m}^2. \quad (3.6)$$

Issiqlik qaynayotgan suvdan tuxum yuzasi orqali tuxumga o'tkaziladi. Bu esa vaqt birligida tuxumga o'tkazilgan energiya miqdori (quvvat) uchun:

$$P = 4\pi R^2 J = 4\pi \kappa R (T_1 - T_0) \approx 19.3 \text{ W}. \quad (3.7)$$

Bundan tuxum markazigacha kerakli issiqlik miqdori o'tishi uchun zarur bo'lgan vaqt τ ni baholaymiz:

$$\tau = \frac{U}{P} = \frac{\mu C R^2 (T_c - T_0)}{3\kappa (T_1 - T_0)} = \frac{16768}{19.3} = 869 \text{ s} \approx 14.5 \text{ min}. \quad (3.8)$$

Chaqmoq (Lightning)

Umumiy zaryad Q – bu rasmdagi egri chiziq ostidagi yuza (area under the curve). Uch-burchak shakli tufayli quyidagini olamiz:

$$Q = \frac{I_0 \tau}{2} = \frac{100 \times 10^3 \cdot 0.1 \times 10^{-3}}{2} = 5 \text{ C.} \quad (3.9)$$

O'rtacha tok (average current):

$$I = \frac{Q}{\tau} = \frac{I_0}{2} = 50 \text{ kA.} \quad (3.10)$$

Bulutning pastki qismi manfiy, yer esa musbat zaryadlanganligi sababli, vaziyat asosan ulgi parallel plastinkali kondensator (giant parallel-plate capacitor) kabi. Chaqmoq sodir bo'lishidan oldin saqlanadigan energiya miqdori $QE_0h/2$ ga teng, bu yerda E_0h – bulut pastki qismi va yer orasidagi kuchlanish farqi (voltage difference) va chaqmoq bu energiyani chiqaradi. Bitta chaqmoq uchun energiya:

$$\frac{QE_0h}{2} = \frac{5 \cdot 300 \times 10^3 \cdot 1000}{2} = 7.5 \times 10^8 \text{ J.}$$

Bundan kelib chiqadiki, siz 100 Vt lampochkani quyidagi muddatga yorita olasiz:

$$t = \frac{32 \times 10^6}{6.5 \times 10^9} \times \frac{7.5 \times 10^8}{100} \approx 10 \text{ soat.} \quad (3.11)$$

Kapillyar tomirlar (Capillary Vessels)

Barcha kapillyarlarni hisobga olgan holda, umumiy gidravlik qarshilik (hydraulic resistance):

$$R_{\text{all}} = \frac{\Delta p}{D} = \frac{1 \times 10^3}{100 \times 10^{-6}} = 10^7 \text{ Pa} \cdot \text{s/m}^3.$$

Barcha kapillyarlar parallel ulangan deb faraz qilinadi. Puazeyl va Om qonunlari o'rtasidagi o'xshashlik (analogy) bitta kapillyarning gidravlik qarshiligi R uchun quyidagini beradi:

$$\frac{1}{R_{\text{all}}} = \frac{N}{R} \Rightarrow N = \frac{R}{R_{\text{all}}}.$$

Endi Puazeyl qonuni yordamida R ni hisoblaymiz:

$$R = \frac{8\eta L}{\pi r^4} = \frac{8 \cdot 4.5 \times 10^{-3} \cdot 1 \times 10^{-3}}{\pi(4 \times 10^{-6})^4} \approx 4.5 \times 10^{16} \text{ kg}/(\text{m}^4 \cdot \text{s}).$$

Shunday qilib, inson tanasidagi kapillyar tomirlar soni:

$$N = \frac{4.5 \times 10^{16}}{10^7} = 4.5 \times 10^9. \quad (3.12)$$

Hajmli oqim (volume flow) $D = S_{\text{all}}v$, bu yerda $S_{\text{all}} = N\pi r^2$ – barcha kapillyar tomirlar bilan bog'liq umumiy ko'ndalang kesim yuzasi. U holda qon tezligi:

$$v = \frac{D}{N\pi r^2} = \frac{100 \times 10^{-6}}{4.5 \times 10^9 \cdot \pi(4 \times 10^{-6})^2} \approx 0.44 \text{ mm/s.} \quad (3.13)$$

(Ikkinchi ifoda bitta kapillyarni alohida ko'rib chiqish orqali ham topiladi: $v = \frac{r^2 \Delta p}{8\eta L}$).

Osmono'par bino (Skyscraper)

Yupqa havo qatlami (slab) yerdan z balandlikda bo'lganda, qatlamdagi havoning bosimi $p(z)$ va harorati $T(z)$ bo'ladi, qatlam hajmi $V(z) = Ah(z)$ (bu yerda A – ko'ndalang kesim yuzasi, $h(z)$ – qatlam qalinligi). Har qanday balandlikda z ideal gaz qonuni (ideal gas law) $pV = NkT$ (N – qatlamdagi molekulalar soni) bilan adiabatik qonun (adiabatic law) $pV^\gamma = \text{const}$ yoki $(pV)^\gamma \propto p^{\gamma-1}$ ni birlashtiramiz va $p^{\gamma-1} \propto T^\gamma$ degan xulosaga kelamiz. Buni differensiallab:

$$\begin{aligned}(\gamma - 1) \frac{dp}{p} &= \gamma \frac{dT}{T} \\ \Rightarrow \frac{dT}{T} &= \left(1 - \frac{1}{\gamma}\right) \frac{dp}{p}.\end{aligned}\quad (3.14)$$

Qatlam tezlanmaganligi sababli, og'irlik (weight) qatlamning ustki va pastki qismidagi bosim farqidan kelib chiqadigan kuch bilan muvozanatlashishi kerak. Pastga yo'nalgan kuchlarni musbat deb olib, aniq kuch:

$$0 = Nmg + A[p(z+h) - p(z)] = \frac{pV}{kT}mg + A\frac{dp}{dz}h.$$

Shunday qilib,

$$\frac{dp}{dz} = -\frac{mg}{k} \frac{p}{T} \quad \text{yoki} \quad dp = -\frac{mg}{k} \frac{p}{T} dz. \quad (3.15)$$

Ikki ifodani birlashtirib:

$$dT = -\left(1 - \frac{1}{\gamma}\right) \frac{mg}{k} dz.$$

Shuning uchun balandligi H bo'lgan bino uchun:

$$T_{\text{top}} = T_{\text{bot}} - \left(1 - \frac{1}{\gamma}\right) \frac{mgH}{k}.$$

$H = 1000$ m, $T_{\text{bot}} = 30^\circ\text{C}$, $\gamma = 7/5$, $m = 4.65 \times 10^{-26}$ kg, $g = 9.80$ m/s², $k = 1.38 \times 10^{-23}$ J/K qiymatlarini qo'yib:

$$T_{\text{top}} = 30 - \left(1 - \frac{5}{7}\right) \cdot \frac{4.65 \times 10^{-26} \cdot 9.80 \cdot 1000}{1.38 \times 10^{-23}} \approx 30 - 0.086 \cdot 1000?$$

To'g'ri hisob:

$$\begin{aligned}\left(1 - \frac{1}{\gamma}\right) &= 1 - \frac{5}{7} = \frac{2}{7} \approx 0.2857. \\ \frac{mgH}{k} &= \frac{4.65 \times 10^{-26} \cdot 9.80 \cdot 1000}{1.38 \times 10^{-23}} = \frac{4.557 \times 10^{-22}}{1.38 \times 10^{-23}} \approx 33.02 \text{ K}. \\ 0.2857 \cdot 33.02 &\approx 9.43 \text{ K}.\end{aligned}$$

Shunday qilib,

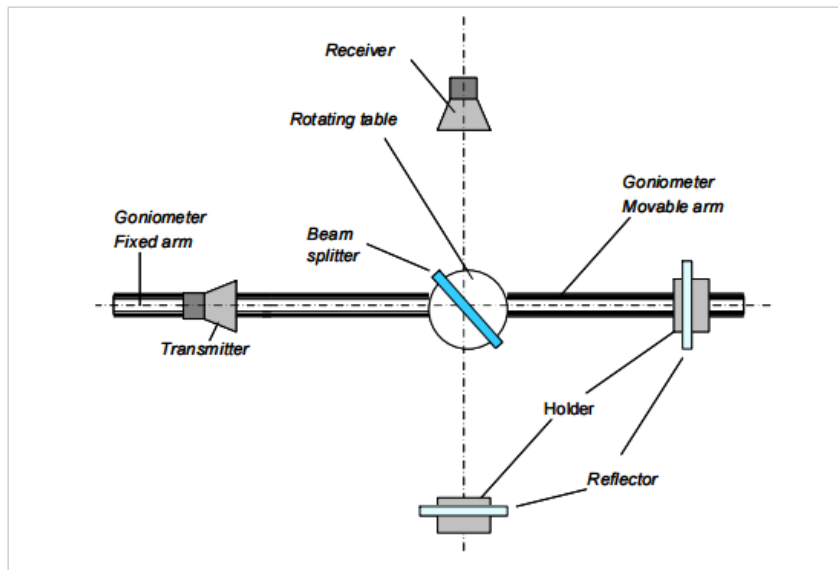
$$T_{\text{top}} = 30^\circ\text{C} - 9.43^\circ\text{C} \approx 20.6^\circ\text{C}. \quad (3.16)$$

The 37th International Physics Olympiad
Singapore
Experimental Competition
Wednesday, 12 July, 2006

Sample Solution

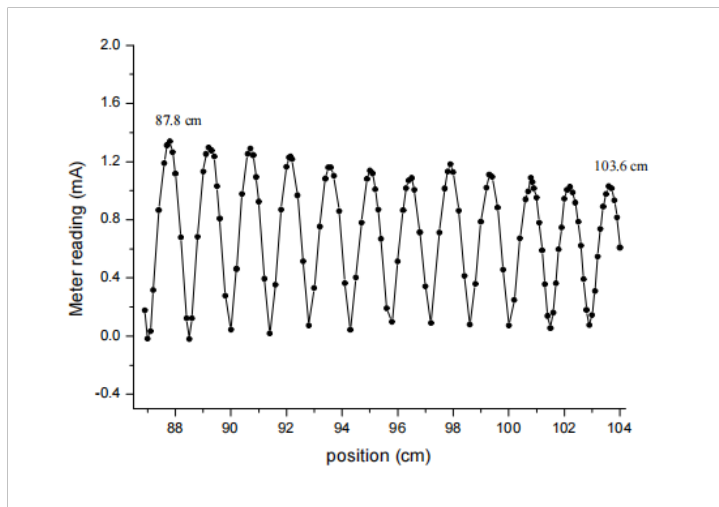
1-qism (davomi)

a. Eksperimental qurilmaning chizmasi (chizma talab qilinmaydi)



Eksperimental qurilma:

Receiver, Rotating table, Goniometer, Fixed arm, Beam splitter, Transmitter, Reflector va boshqa komponentlar.



Grafik (grafik talab qilinmaydi)

yoki boshqa usul bilan birinchi maksimum nuqta 87.8 sm va 12-maksimum nuqta 103.6 sm da o'lchangan.

To'lqin uzunligi (wavelength) quyidagicha hisoblanadi:

$$\frac{\lambda}{2} = \frac{103.6 - 87.8}{11} \text{ sm}$$

Shunday qilib,

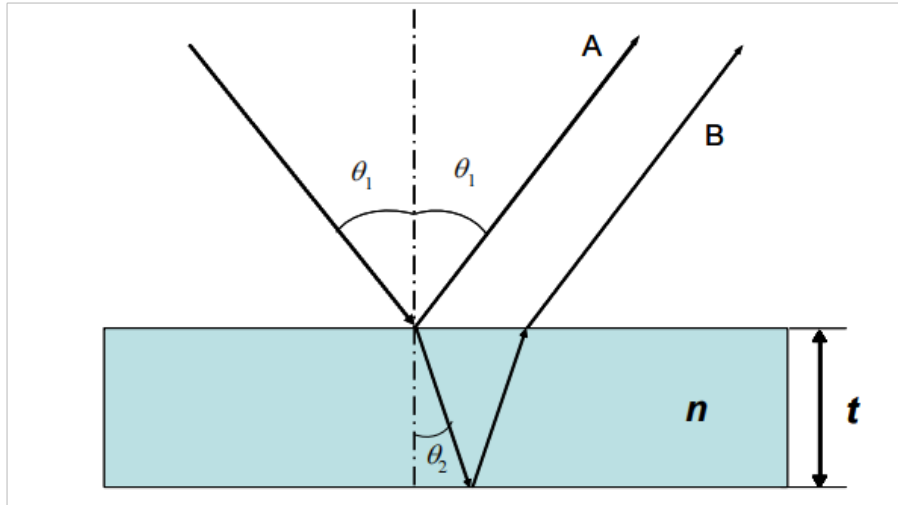
$$\lambda = 2.87 \text{ sm. (1.8 ball)}$$

Xatolik tahlili (Error analysis)

$$\lambda = \frac{2}{11}d, \quad \Delta d = 0.05 \times 2 \text{ sm} = 0.1 \text{ sm.}$$

$$|\Delta\lambda| = \left| \frac{2}{11} \Delta d \right| = \frac{2}{11} \times 0.10 = 0.018 \text{ sm} < 0.02 \text{ sm} \quad (0.2 \text{ ball})$$

2-qism: “Yupqa plyonka” interferensiyasi



(a) Interferensiya shartlarini keltirib chiqarish (Deduction of interference conditions)

Plyonkaning qalinligi t va sindirish ko'rsatkichi n deb faraz qilaylik. θ_1 – tushish burchagi, θ_2 – sinish burchagi (refracted angle). Optik yo'llar farqi ΔL :

$$\Delta L = 2 \left(\frac{nt}{\cos \theta_2} - t \tan \theta_2 \sin \theta_1 \right)$$

Sinish qonuni (law of refraction):

$$\sin \theta_1 = n \sin \theta_2$$

Shunday qilib,

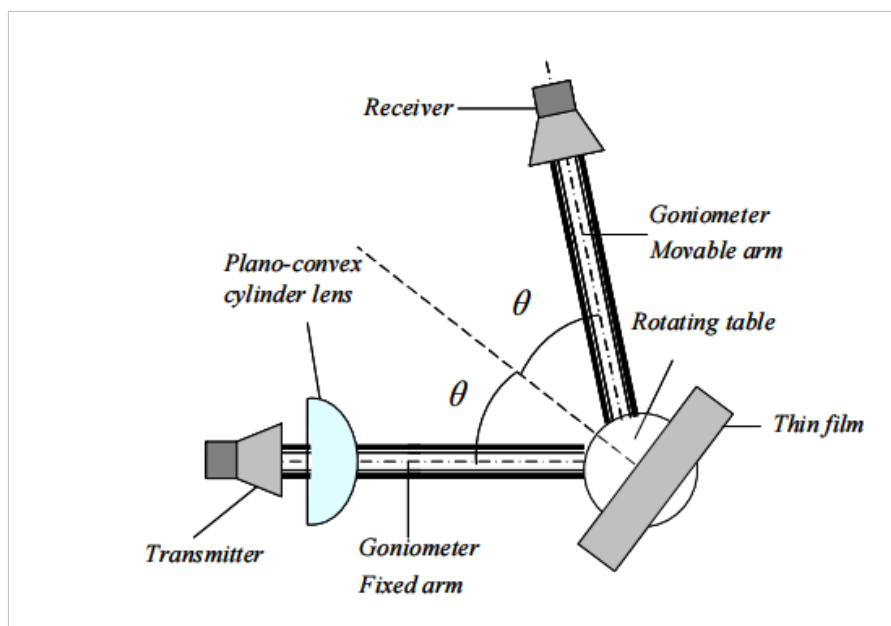
$$\Delta L = 2t \sqrt{n^2 - \sin^2 \theta_1}$$

Havo-plyonka chegarasida aks etgan nurda 180° (π) faza siljishi (phase shift) ni hisobga olgan holda, interferensiya shartlari:

$$2t \sqrt{n^2 - \sin^2 \theta_{\min}} = m\lambda \quad (m = 1, 2, 3, \dots) \quad \text{destruktiv cho'qqi (minimum) uchun}$$

$$2t \sqrt{n^2 - \sin^2 \theta_{\max}} = \left(m \pm \frac{1}{2} \right) \lambda \quad \text{konstruktiv cho'qqi (maksimum) uchun}$$

Agar qalinlik t va to'lqin uzunligi λ ma'lum bo'lsa, $I-\theta_1$ spektridan (I – interferensiyalashgan nurning intensivligi) yupqa plyonkaning sindirish ko'rsatkichini aniqlash mumkin.



Eksperimental qurilma chizmasi (talabalar 2-sahifadagi yorliqlardan foydalanishlari kerak).

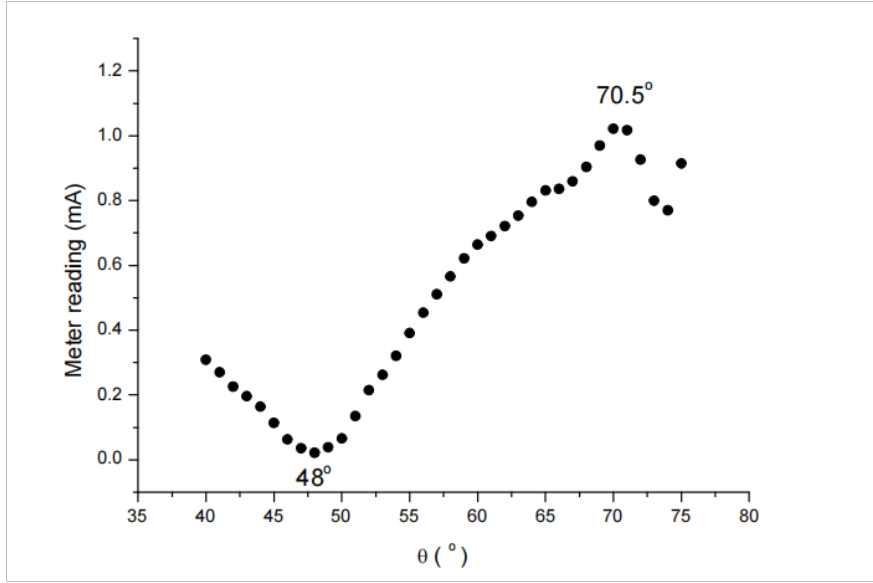
(b) va (c) Ma'lumotlar to'plami (Data Set)

X: θ_i (daraja)	Y: Metr ko'rsatkichi S (mA)
40.0	0.309
41.0	0.270
42.0	0.226
43.0	0.196
44.0	0.164
45.0	0.114
46.0	0.063
47.0	0.036
48.0	0.022
49.0	0.039
50.0	0.066
51.0	0.135
52.0	0.215
53.0	0.262
54.0	0.321
55.0	0.391
56.0	0.454
57.0	0.511
58.0	0.566
59.0	0.622
60.0	0.664
61.0	0.691
62.0	0.722
63.0	0.754
64.0	0.796
65.0	0.831
66.0	0.836
67.0	0.860
68.0	0.904
69.0	0.970
70.0	1.022
71.0	1.018
72.0	0.926
73.0	0.800
74.0	0.770
75.0	0.915

Noaniqlik (Uncertainty): burchak $\Delta\theta_1 = \pm 0.5^\circ$, tok (current): ± 0.001 mA.

Ma'lumotlardan $\theta_{\min}$ va $\theta_{\max}$ mos ravishda 48° va 70.5° da topiladi.

Sindirish ko'rsatkichini (refractive index) hisoblash uchun quyidagi tenglamalar qo'llaniladi:



3-qism eksperimental qurilmasi (talabalar 2-sahifadagi yorliqlardan foydalanishlari kerak).

$$2t\sqrt{n^2 - \sin^2 48^\circ} = m\lambda \quad (m = 1, 2, 3, \dots) \quad (1)$$

va

$$2t\sqrt{n^2 - \sin^2 70.5^\circ} = \left(m - \frac{1}{2}\right) \lambda \quad (2)$$

Ushbu tajribada $t = 5.28$ sm, $\lambda = 2.85$ sm (boshqa usul bilan o'lchangan).

(1) va (2) tenglamalar sistemasini yechib:

$$m = \frac{\sin^2 70.5^\circ - \sin^2 48^\circ}{\left(\frac{\lambda}{2t}\right)^2} + 0.25$$

$$m = 4.83 \longrightarrow m = 5$$

$m = 5$ ni (1) ga qo'yib, $n = 1.54$ olamiz. $m = 5$ ni (2) ga qo'yib ham $n = 1.54$ olamiz.

Xatolik tahlili (Error analysis)

$$n = \sqrt{\sin^2 \theta + \left(\frac{m\lambda}{2t}\right)^2}$$

$$\Delta n = \frac{1}{\sqrt{\sin^2 \theta + \left(\frac{m\lambda}{2t}\right)^2}} \left(\sin 2\theta \cdot \Delta \theta + \frac{m^2 \lambda}{2t^2} \Delta \lambda - \frac{m^2 \lambda^2}{2t^3} \Delta t \right)$$

$$= \frac{1}{n} \left(\sin 2\theta \cdot \Delta \theta + \frac{m^2 \lambda}{2t^2} \Delta \lambda - \frac{m^2 \lambda^2}{2t^3} \Delta t \right)$$

Agar $\Delta \theta = \pm 0.5^\circ = \pm 0.0087$ rad, $\Delta t = \pm 0.05$ sm, $\Delta \lambda = \pm 0.02$ sm va $\theta = 48^\circ$ olinsa:

$$\Delta n = \frac{1}{1.54} \left(0.0087 \sin 96^\circ + \frac{5^2 \times 2.85}{2 \times 5.28^2} \times 0.01 + \frac{5^2 \times 2.85^2}{2 \times 5.28^3} \times 0.05 \right) \approx 0.02$$

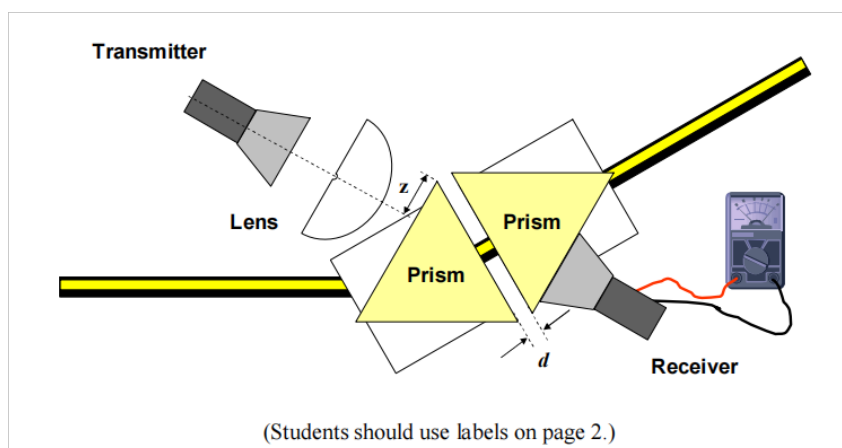
Shunday qilib,

$$\boxed{n + \Delta n = 1.54 \pm 0.02}$$

3-qism (Part 3): Namunaviy yechim (Sample Solution)

1-topshiriq (Task 1)

Yakuniy eksperimental qurilmani chizing va 2-sahifadagi yorliqlardan foydalanib barcha komponentlarni belgilang. Chizmangizda z masofasini (3.2-rasmga qarang) yozing, bu yerda z – prizma uchidan uzatgichning markaziy o'qigacha bo'lgan masofa.



Eksperimental qurilma:

Transmitter, Lens, Prism, Prism, Receiver. (Talabalar 2-sahifadagi yorliqlardan foydalanishlari kerak.)

2-topshiriq (Task 2)

Ma'lumotlaringizni jadvalga yozing. Tajribani ikki marta bajaring.

Ma'lumotlar to'plami (Data Set)

X: d (sm) ΔY &	ΔX (sm)	Set 1	Set 2	$S_{o'rtacha}$ (mA)	ΔS (mA)#	I_t (mA ²)*	ΔI_t §	Y: $\ln(I_t)$
0.60 0.03	0.05	0.78	0.78	0.780	0.01	0.6080	0.016	-0.50
0.70 0.03	0.05	0.68	0.69	0.685	0.01	0.4690	0.014	-0.76
0.80 0.03	0.05	0.58	0.59	0.585	0.01	0.3420	0.012	-1.07
0.90 0.04	0.05	0.50	0.51	0.505	0.01	0.2550	0.010	-1.37
1.00 0.05	0.05	0.42	0.42	0.420	0.01	0.1760	0.008	-1.74
1.10 0.06	0.05	0.36	0.35	0.355	0.01	0.1260	0.007	-2.07
1.20 0.06	0.05	0.31	0.31	0.310	0.01	0.0961	0.006	-2.34
1.30 0.08	0.05	0.26	0.25	0.255	0.01	0.0650	0.005	-2.73
1.40 0.09	0.05	0.21	0.22	0.215	0.01	0.0462	0.004	-3.07

$\Delta S = 0.01$ mA (har bir tok o'lchovlari to'plami uchun)

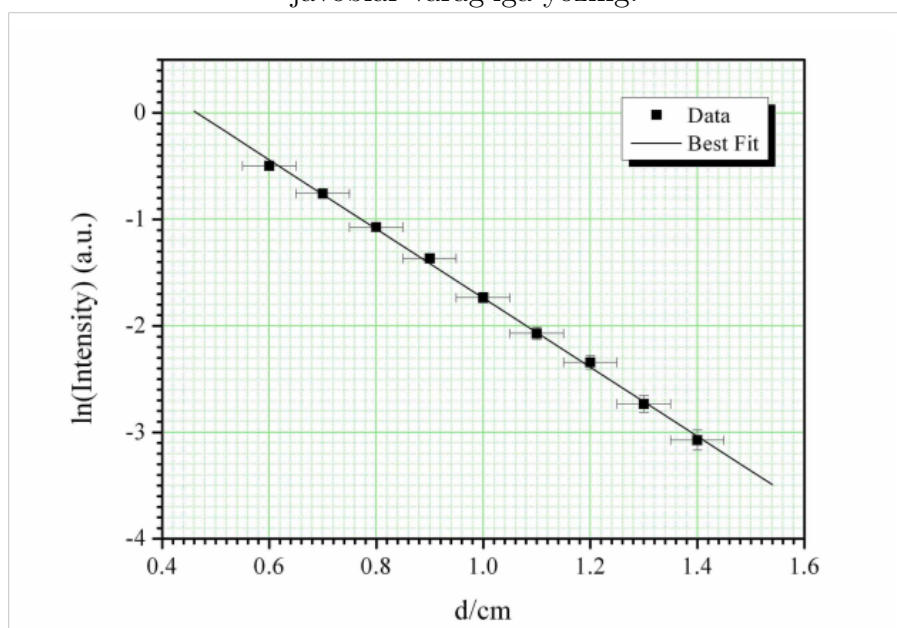
* S^2 intensivlik I_t ga proporsional

§ $\Delta(S^2) = \Delta I_t = 2S \times \Delta S$

& $\Delta Y = \Delta(\ln I_t) = \Delta I_t / I_t$

3-topshiriq (Task 3)

Tegishli grafiklarni chizish orqali prizmaning sindirish ko'rsatkichi n_1 ni xatolik tahlili bilan aniqlang. Prizmaning sindirish ko'rsatkichi n_1 va uning noaniqligi Δn_1 ni berilgan javoblar varag'iga yozing.



Eng kichik kvadratlar usuli (Least Square Fitting)

$X = d \text{ (sm)}$	$\Delta X \text{ (sm)}$	$Y = \ln(I_t)$	ΔY	ΔY^2	XY	X^2
0.60	0.05	-0.50	0.03	0.001	-0.298	0.36
0.70	0.05	-0.76	0.03	0.001	-0.530	0.49
0.80	0.05	-1.07	0.03	0.001	-0.858	0.64
0.90	0.05	-1.37	0.04	0.002	-1.230	0.81
1.00	0.05	-1.74	0.05	0.002	-1.735	1.00
1.10	0.05	-2.07	0.06	0.003	-2.278	1.21
1.20	0.05	-2.34	0.06	0.004	-2.811	1.44
1.30	0.05	-2.73	0.08	0.006	-3.553	1.69
1.40	0.05	-3.07	0.09	0.009	-4.304	1.96
$\Sigma X = 9.00$	$\Sigma Y = -15.648$	$\Sigma \Delta Y = 0.469$	$\Sigma (\Delta Y)^2 = 0.029$	$\Sigma XY = -17.596$	$\Sigma X^2 = 9.600$	$\Sigma Y^2 = 15.648$

$I_t = I_0 \exp(-2\gamma d)$ dan ikkala tomonning natural logarifmini olib:

$$\ln(I_t) = -2\gamma d + \ln(I_0)$$

Bu $y = mx + c$ ko'rinishida.

Gradientni (slope) hisoblash uchun quyidagi tenglama qo'llanildi, bu yerda N – ma'lumot nuqtalari soni:

$$m = \frac{N \Sigma(XY) - (\Sigma X)(\Sigma Y)}{N \Sigma X^2 - (\Sigma X)^2} = -3.247$$

Individual Y ma'lumot qiymatlarining standart og'ishi σ_Y ni hisoblash uchun:

$$\sigma_Y = \sqrt{\frac{\Sigma(\Delta Y)^2}{N - 2}} = 0.064$$

Shunday qilib, qiyalikning standart og'ishi (standard deviation in the slope):

$$\sigma_m = \sigma_Y \sqrt{\frac{N}{\Sigma X^2 - (\Sigma X)^2}} = 0.082$$

Gradientdan:

$$2\gamma = 3.247 \pm 0.082 \approx 3.25 \pm 0.08$$

Quyidagi formuladan foydalanib:

$$n_1 = \frac{\sqrt{k_2^2 + \gamma^2}}{k_2 \sin \theta_1}$$

bu yerda $\theta_1 = 60^\circ$, $k_2 = 2\pi/\lambda \approx 2.20$ (oldingi qismdan aniqlangan to'lqin uzunligi $\lambda = (2.85 \pm 0.02)$ sm yordamida), biz quyidagini olamiz:

$$n_1 \pm \Delta n_1 = 1.434 \pm 0.016 \approx 1.43 \pm 0.02$$

n_1 sindirish ko'rsatkichi uchun xatolik tahlili (Error analysis for refractive index of n_1)

$$\begin{aligned}\Delta n_1 &= \frac{d}{dk_2} \left[\left(\frac{k_2^2 + \gamma^2}{k_2 \sin \theta_1} \right)^{1/2} \right] \Delta k_2 + \frac{d}{d\gamma} \left[\left(\frac{k_2^2 + \gamma^2}{k_2 \sin \theta_1} \right)^{1/2} \right] \Delta \gamma \\ &= \left[\frac{k_2}{\sqrt{k_2^2 + \gamma^2} \cdot k_2 \sin \theta_1} - \frac{\sqrt{k_2^2 + \gamma^2}}{k_2^2 \sin \theta_1} \right] \Delta k_2 + \left[\frac{\gamma}{\sqrt{k_2^2 + \gamma^2} \cdot k_2 \sin \theta_1} \right] \Delta \gamma \\ &\approx 0.016 \approx 0.02\end{aligned}$$

bu yerda:

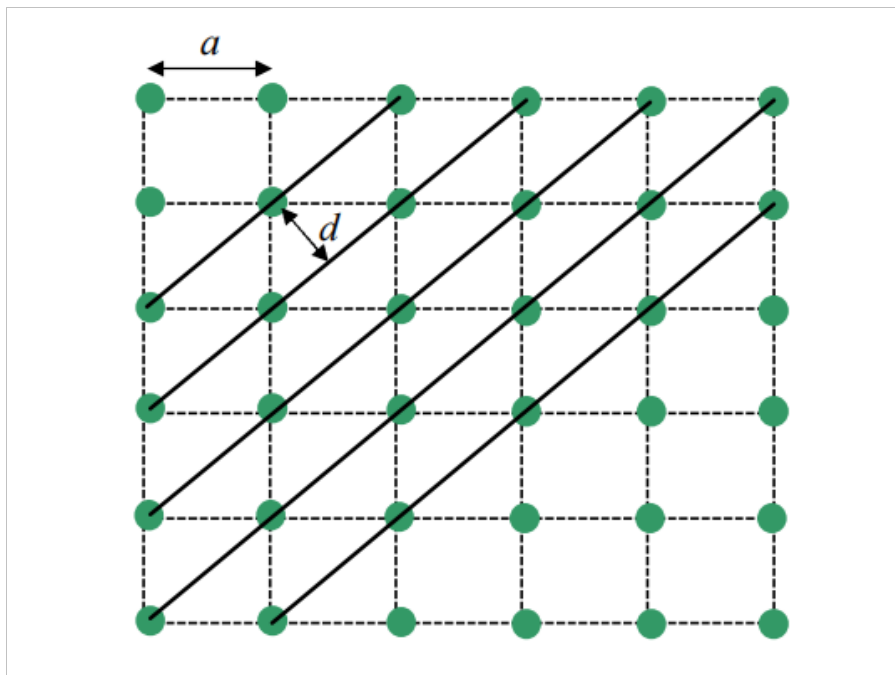
$$\Delta k_2 = -\frac{2\pi}{\lambda^2} \Delta \lambda = -0.015$$

Eslatma: Xatolik tahlilining boshqa usullari ham qabul qilinadi.

4-qism: Metall-rod panjarasining mikroto'lqin diffraksiyasi (davomi)

1-topshiriq (Task 1)

Oddiy kvadrat panjaraning ustki ko'rinishi (top-view of a simple square lattice).



4.1-rasm: Panjara konstantasi a va diagonal tekisliklarning tekisliklararo masofasi (interplanar spacing) d ko'rsatilgan oddiy kvadrat panjaraning sxematik diagrammasi.

Bragg qonunini keltirib chiqarish (Deriving Bragg's Law)

Diffraksiya cho'qqilarini (diffraction peaks) kuzatish uchun zarur shartlar:

1. Tushish burchagi (angle of incidence) = sochilish burchagi (angle of scattering).
2. Yo'l uzunliklari farqi (path length difference) to'lqin uzunliklarining butun soniga teng.

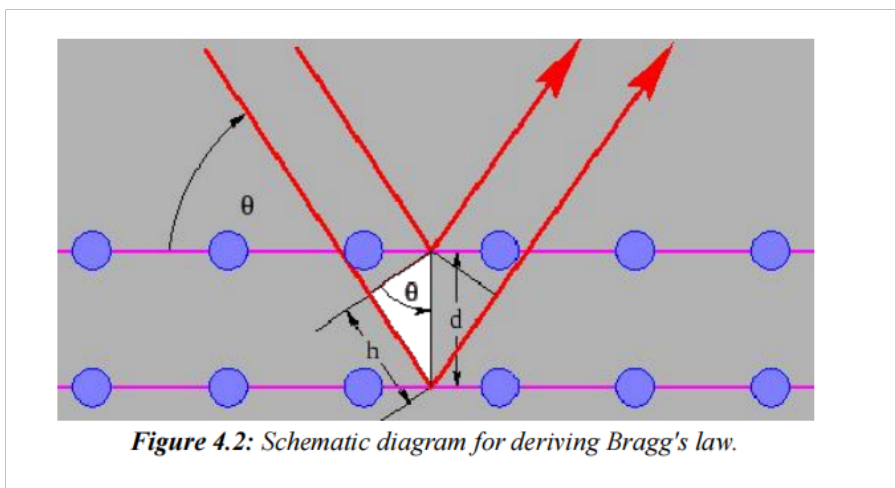


Figure 4.2: Schematic diagram for deriving Bragg's law.

4.2-rasm: Bragg qonunini keltirib chiqarish uchun sxematik diagramma.

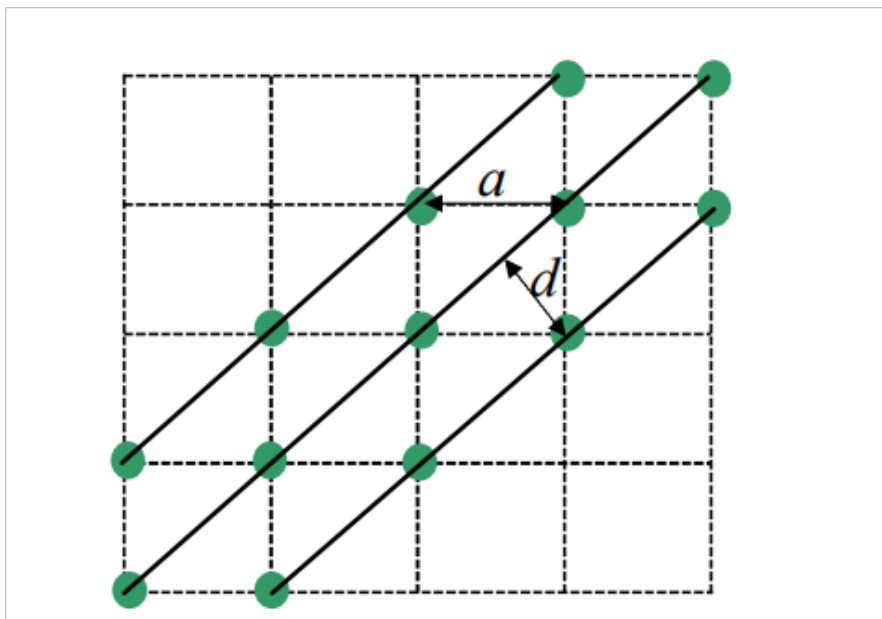
$$h = d \sin \theta \quad (1)$$

Yo'l uzunliklari farqi:

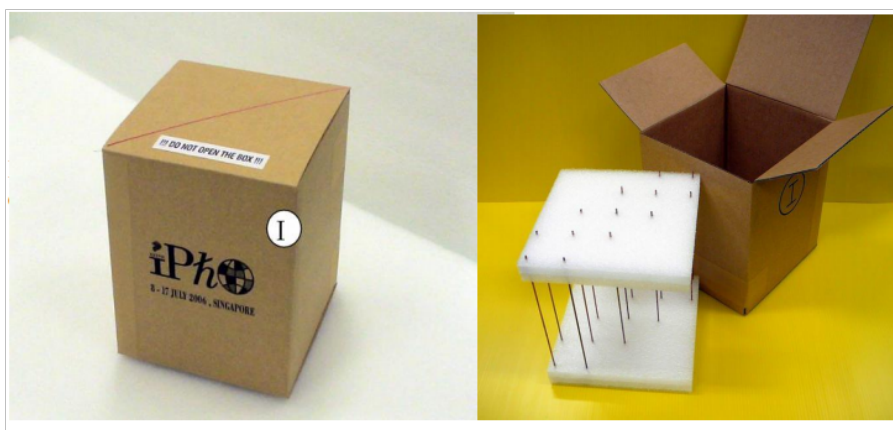
$$2h = 2d \sin \theta \quad (2)$$

Diffraktsiya sodir bo'lishi uchun yo'l farqi quyidagini qanoatlantirishi kerak:

$$2d \sin \theta = m\lambda, \quad m = 1, 2, 3, \dots \quad (3)$$

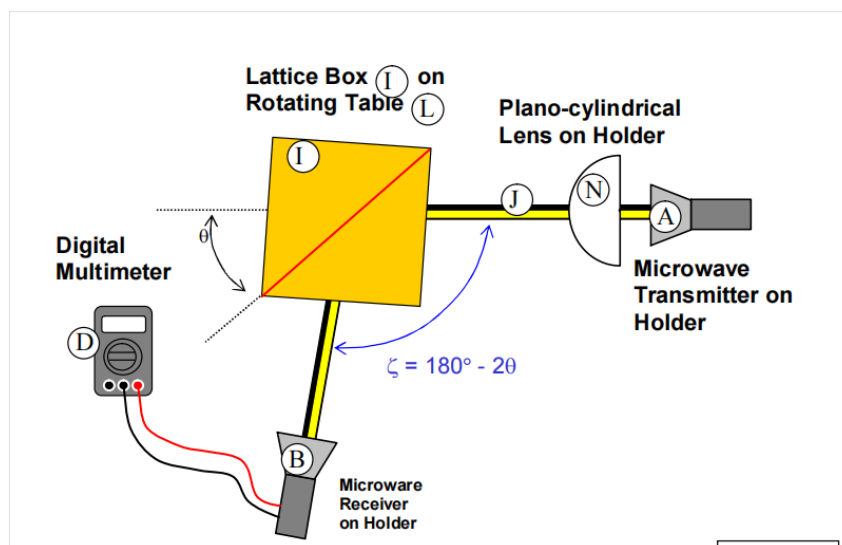


4.3-rasm: Tajribada ishlatilgan panjaraning tasviri (bu rasm talab qilinmaydi).



4.4-rasm: Tajribada ishlatilgan haqiqiy panjara (bu rasm talab qilinmaydi).

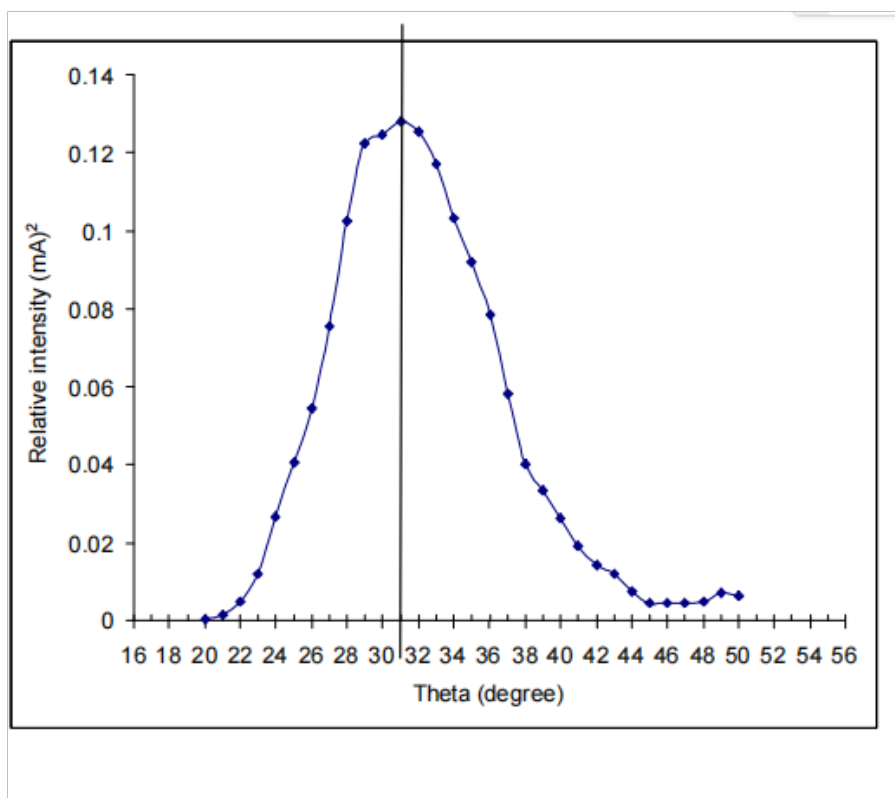
2-topshiriq (2(a))



4.5-rasm: Eksperimental qurilma chizmasi. Lattice Box on Rotating Table, Plano-cylindrical Lens on Holder, Digital Multimeter, Microwave Transmitter on Holder, Microwave Receiver on Holder. $\zeta = 180^\circ - 2\theta$

2-topshiriq (2(b) va 2(c)) – Ma'lumotlar to'plami (Data Set)

$\theta/^\circ$	$\zeta/^\circ$	Chiqish toki S (mA)	Intensivlik $I = S^2$ (mA ²)
20.0	140.0	0.023	0.000529
21.0	138.0	0.038	0.001444
22.0	136.0	0.070	0.0049
23.0	134.0	0.109	0.011881
24.0	132.0	0.163	0.026569
25.0	130.0	0.201	0.040401
26.0	128.0	0.233	0.054289
27.0	126.0	0.275	0.075625
28.0	124.0	0.320	0.1024
29.0	122.0	0.350	0.1225
30.0	120.0	0.353	0.124609
31.0	118.0	0.358	0.128164
32.0	116.0	0.354	0.125316
33.0	114.0	0.342	0.116964
34.0	112.0	0.321	0.103041
35.0	110.0	0.303	0.091809
36.0	108.0	0.280	0.0784
37.0	106.0	0.241	0.058081
38.0	104.0	0.200	0.04
39.0	102.0	0.183	0.033489
40.0	100.0	0.162	0.026244
41.0	98.0	0.139	0.019321
42.0	96.0	0.120	0.0144
43.0	94.0	0.109	0.011881
44.0	92.0	0.086	0.007396
45.0	90.0	0.066	0.004356
46.0	88.0	0.067	0.004489
47.0	86.0	0.066	0.004356
48.0	84.0	0.070	0.0049
49.0	82.0	0.084	0.007056
50.0	80.0	0.080	0.0064



2-topshiriq (2(d)) – Panjara konstantasini aniqlash

(3) tenglamadan $m = 1$ deb va $\theta_{\max}$ – diffraktsiya maksimumiga mos keladigan burchak:

$$2d \sin \theta_{\max} = \lambda$$

4.3-rasmdan (yoki o'lchangan grafikdan) $\theta_{\max}$ aniqlanadi va λ oldingi qismdan ma'lum. Shunday qilib:

$$d = \frac{\lambda}{2 \sin \theta_{\max}}$$

4.3-rasmga ko'ra, a va d orasidagi bog'lanish: $d = \frac{a}{\sqrt{2}}$ (kvadrat panjara diagonali bo'yicha). Shuning uchun:

$$a = \sqrt{2}d = \frac{\lambda\sqrt{2}}{2 \sin \theta_{\max}}$$

4-qism (davomi) – Panjara konstantasini hisoblash

(4) va (5) tenglamalarni birlashtirib, quyidagini olamiz:

$$a = \frac{\lambda}{\sqrt{2} \sin \theta_{\max}}$$

Ma'lumotlarning simmetriyasidan cho'qqi (peak) joylashuvi aniqlanadi:

$$\theta_{\max} = 31^\circ \quad (\text{Nazariy qiymat } \theta_{\max} = 32^\circ)$$

Shunday qilib,

$$a = \frac{\lambda}{\sqrt{2} \sin \theta_{\max}} = \frac{2.85 \text{ sm}}{\sqrt{2} \sin 31^\circ} = 3.913 \text{ sm}$$

(Haqiqiy qiymat $a = 3.80$ sm).

Baholash sxemasidagi 3.55 qiymati quyidagicha olingan:

$$a = \frac{\lambda}{\sqrt{2} \sin \theta_{\max}} = \frac{2.83 \text{ sm}}{\sqrt{2} \sin 34^\circ} = 3.58 \text{ sm}$$

bu yerda 2.83 sm va 34° – to'lqin uzunligi va cho'qqi joylashuvi uchun ruxsat etilgan minimal va maksimal qiymatlar.

Xuddi shunday:

- 4.10 qiymati: $a = \frac{2.87}{\sqrt{2} \sin 30^\circ} = 4.06 \text{ sm}$
- 3.55 qiymati: $a = \frac{2.83}{\sqrt{2} \sin 34^\circ} = 3.58 \text{ sm}$
- 3.40 qiymati: $a = \frac{2.83}{\sqrt{2} \sin 35^\circ} = 3.49 \text{ sm}$
- 4.20 qiymati: $a = \frac{2.87}{\sqrt{2} \sin 29^\circ} = 4.18 \text{ sm}$

Xatolik tahlili (Error analysis)

Ma'lum noaniqliklar (uncertainties):

$$\Delta\lambda = 0.02 \text{ sm}, \quad \Delta\theta = 0.5^\circ = 0.014 \text{ rad}$$

(θ ni grafikdan aniqlashdagi noaniqlik).

$$a = \frac{\lambda}{\sqrt{2} \sin \theta_{\max}}$$

dan:

$$\begin{aligned} \Delta a &= \frac{\Delta\lambda}{\sqrt{2} \sin \theta_{\max}} - \frac{\lambda}{\sqrt{2} (\sin \theta_{\max})^2} \cdot \frac{d}{d\theta} (\sin \theta_{\max}) \Delta\theta \\ &= a \left(\frac{\Delta\lambda}{\lambda} - \frac{1}{\sin \theta_{\max}} \cdot \frac{d}{d\theta} (\sin \theta_{\max}) \Delta\theta \right) \\ &= a \left(\frac{\Delta\lambda}{\lambda} - \cot \theta_{\max} \cdot \Delta\theta \right) \end{aligned}$$

Raqamli qiymatlarni qo'yib:

$$\begin{aligned} \Delta a &= 3.80 \left(\frac{0.02}{2.85} - \cot 32^\circ \times (-0.014) \right) \text{ sm} \\ &= 3.80 (0.00702 + 0.0224) \approx 0.112 \text{ sm} \approx 0.1 \text{ sm} \end{aligned}$$

Shunday qilib,

$$a \pm \Delta a = 3.913 \pm 0.112 \approx 3.9 \pm 0.1 \text{ sm}$$

(0.8 ball)